

PLAYER AND MODDER GUIDE

STRATA

An isometric block world, with a Minecraft and Forge compatibility layer

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Covers Strata v0.34.0 and the **mineCompatibility** branch (Android build 45 onward)

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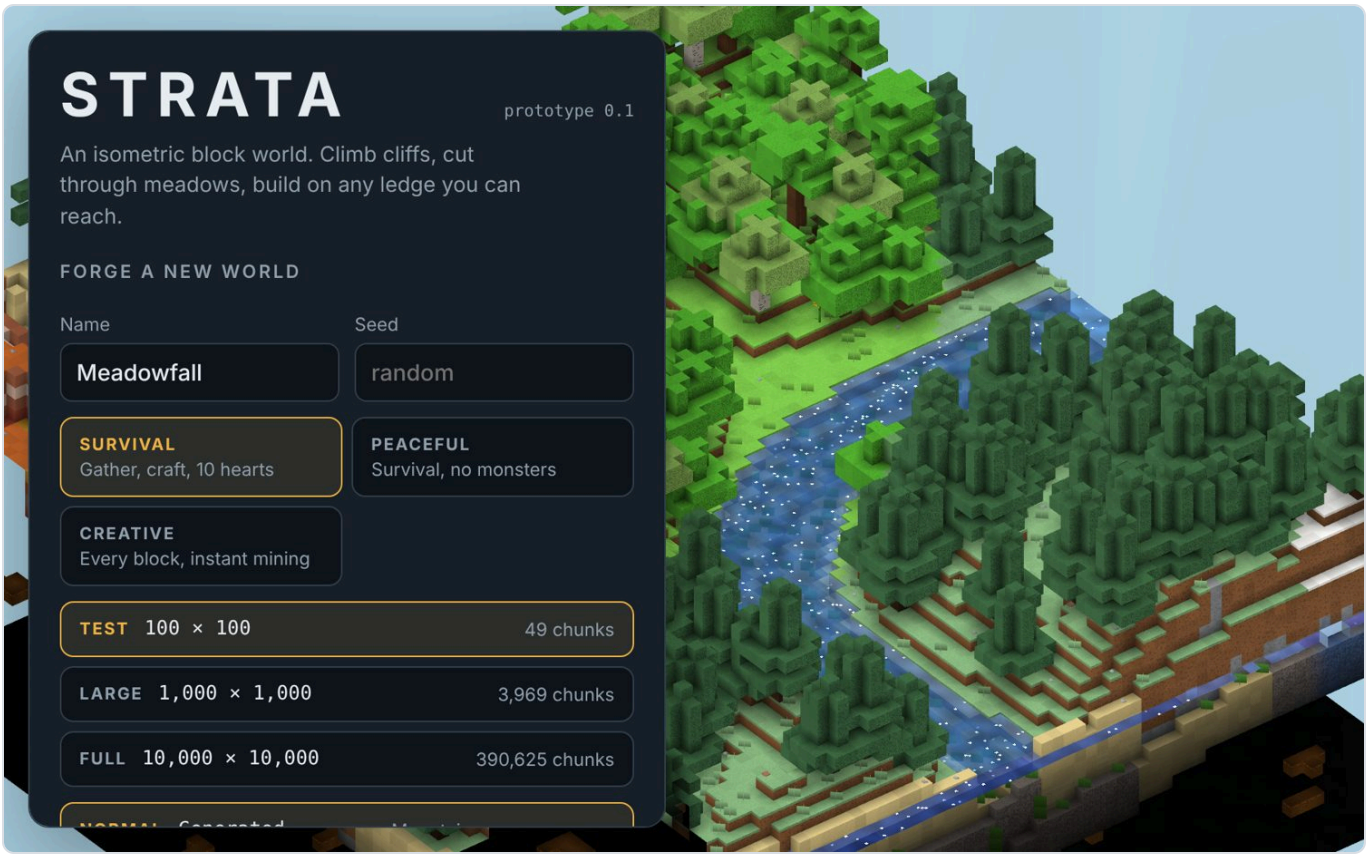
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What Strata is, and what to expect

Strata is a 2.5D isometric voxel sandbox that runs as a single HTML5 file, built to play well on a phone. It has its own survival game, tech and magic, and it can also join real Minecraft: Java Edition servers through an optional compatibility layer.



The title screen: name a world, pick Survival, Peaceful or Creative, a size (100, 1,000 or 10,000 blocks across), a world type and a depth. Multiplayer is at the bottom.

Features at a glance

Worlds

Seeded worlds streamed in chunks; Mini (64 tall), Classic (128) and Full (256) depths; caves, ravines, rivers, oceans, about 30 biomes; a flat Showcase world with every block on display.

Survival

Tools through eleven material tiers, smelters and alloys, armour, farming, cooking, fishing, XP, grimoires (enchanting) and an anvil.

Movement

Climbing with stamina, sprint, dash, swimming, gliding, boats, horses and minecarts. Iso, over-the-shoulder and first-person cameras.

Tech

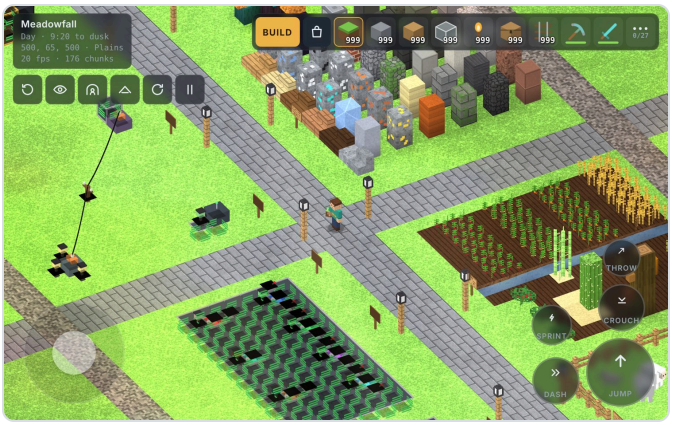
A foundry, RF power, windmills, solar panels I–VIII, logic gates, lamps, silicon chips, wireless terminals, pistons, pipes and hoppers.

Magic

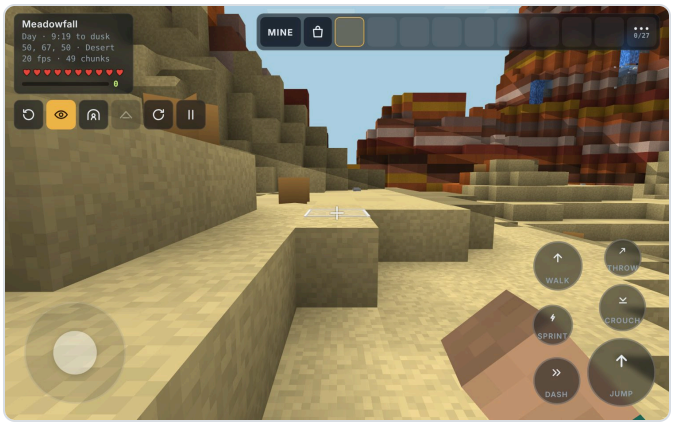
Five kinds of mage that drop essences, a witch's cauldron, witches glass, robes, potions and staffs.

Worlds beyond

The Underworld (8:1 scale, lava seas), the Dusk (Lost villages, star-island libraries) and pocket dimensions.



The Showcase world in Creative: farms, an arboretum of all eleven woods, a stone gallery and machines laid out to explore.



First-person view (the eye button). The game is designed around the isometric view, but every view is playable.

Honest expectations

- **It's a prototype that's actively growing.** Things change between builds, and some features are first versions. Report odd behaviour with a screenshot; most fixes land quickly.
- **Saves live on your device.** Single-player worlds are kept in the browser's (or the app's) local storage. Clearing site data deletes them, and they don't sync between devices.
- **Phones first.** Everything works by touch, in portrait and landscape. A keyboard works too. Big worlds and long view distances need a reasonably modern phone.
- **Minecraft compatibility is a client, not Minecraft.** Strata draws Minecraft servers with its own engine, so some things look or feel different, and things it has no model for are skipped. Chapter 7 lists what to expect.
- **Forge mods need a port.** Any mod's blocks and items show up, but a mod's special screens and behaviour only work when Strata has a port for it. Chapter 8 has the list.
- **No Minecraft files are included.** Without a resource pack, Minecraft servers are drawn with Strata's own textures. Chapter 10 shows how to make a pack from your own game files.

Controls

Strata is played by touch first. Every on-screen button has a keyboard equivalent.

Touch layout



The landscape layout (a 100 × 100 survival world). Portrait uses the same buttons, rearranged.

Control	What it does
Joystick (bottom left)	Walk; push it all the way to run. Movement follows the screen, so up is always "away from you".
Jump	Jump; at a wall, start climbing (uses stamina). On an elevator: go up one floor. In the air with a glider (or an elytra on a server): glide.
Crouch	Toggle crouching (sneak). On an elevator: go down. On Minecraft servers, crouch + tap is the "sneak + right click" many mods use.
Sprint / Dash	Sprint toggles fast movement; Dash is a quick burst.
Throw	Drop one of the held item per tap; hold to drop the whole stack.
Mine / Build	Switches what a tap on a block does: break it, or place or use the held item.
Bag / •••	Inventory and crafting / the full inventory.

Control	What it does
Camera row	Rotate left, first-person eye, "close follow when hidden", camera tilt, rotate right, pause menu.
Two fingers	Pinch to zoom the camera.
Chat and players	Only on Minecraft servers: open chat, and show the online player list.

Keyboard and mouse

Key	Action	Key	Action
`W` `A` `S` `D` / arrows	Move	`Shift`	Run
`Space`	Jump / climb / go up an elevator	`C` or `Ctrl`	Crouch toggle / go down an elevator
`F`	Dash	`B`	Toggle Mine / Build
`Q` / `E`	Rotate camera left / right (`Q` also drops the held item; with `Ctrl` or `Shift`, the stack)	`T`	Change camera tilt
`V`	First-person view	`H`	Close follow when hidden (cave view)
`1`–`9`	Select a hotbar slot	`I`	Inventory
`Esc`	Pause menu / close a screen	Mouse wheel	Zoom (outside first person)

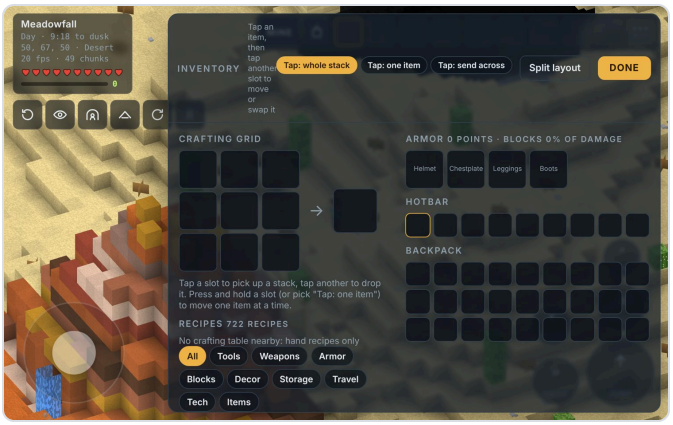
Taps in the world and in inventories

In the world

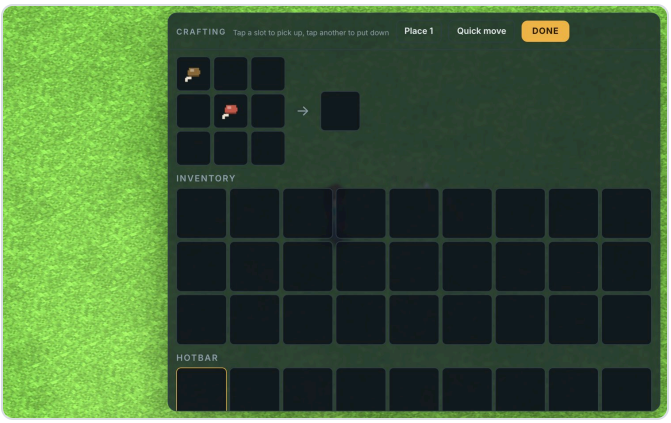
- **Tap a block**: in Mine mode, break it (hold to keep mining); in Build mode, place the held block or use the held item on it (doors, chests, machines, tape on a chest…).
- **Tap the air**: use the held item (eat, drink, throw, bow, books, mod items such as warp pages).
- **Hold** with food or drink: eat or drink, with the animation.
- **Tap a creature**: attack it in Mine mode; in Build mode, interact (ride, trade, feed, shear).

In inventories

- **Tap mode buttons** at the top of the window: **Stack** (a tap picks up the whole stack), **One** (one item at a time) and **Send across** (a tap moves the stack to the other side, like shift-click).
- Tap a slot to pick up, tap another to put down. Press and hold to take one.
- On Minecraft servers in creative, the **Creative** toggle shows a searchable item list beside your inventory, including modded items.



Strata's own inventory: crafting grid, armour, hotbar, backpack and a recipe list sorted by category.



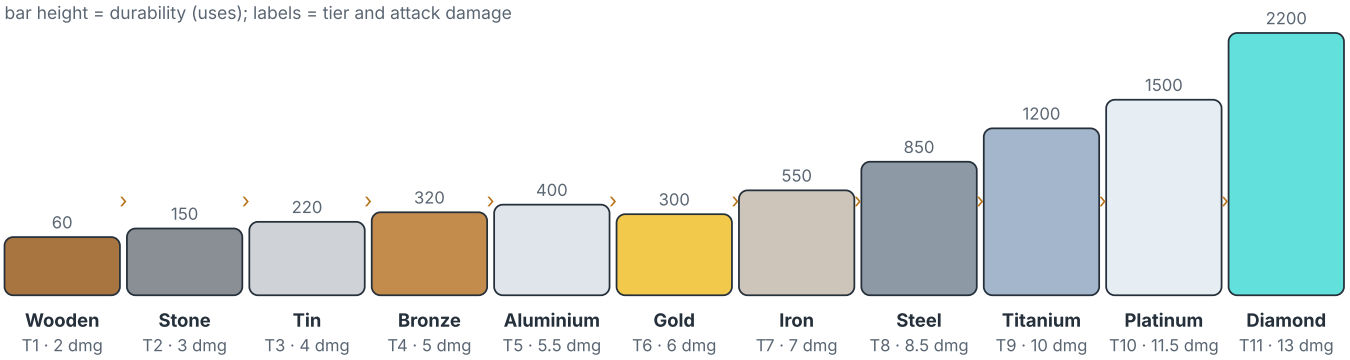
A server's crafting table in compatibility mode, with the same tap modes.

Strata's tech tree

Strata's progression is its own: eleven material tiers for tools and tables, a foundry-to-wireless machine tree, and a separate magic branch. None of it depends on Minecraft or the compatibility layer.

Material tiers

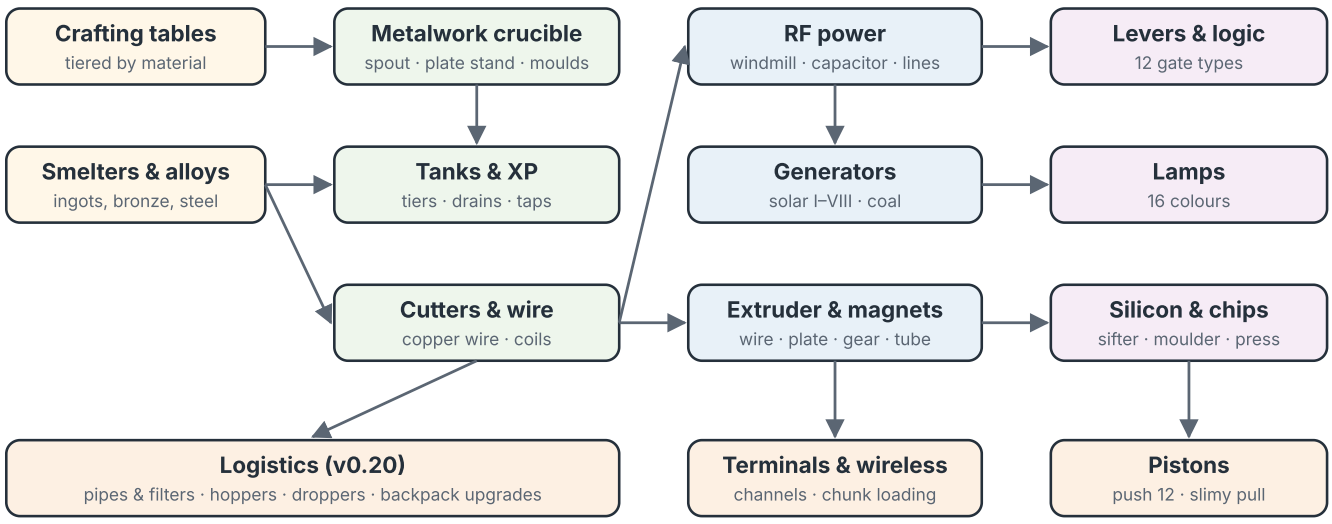
Each material sets which crafting tables can make it, which ores its pickaxe can break, how long it lasts and how fast it works. The tools in every tier are the pickaxe, axe, shovel, hammer and knife, plus weapons, hoes and shears.



Durability and damage are the game's own numbers. Gold is soft but fast; tin and bronze come before iron, as in early metalworking.

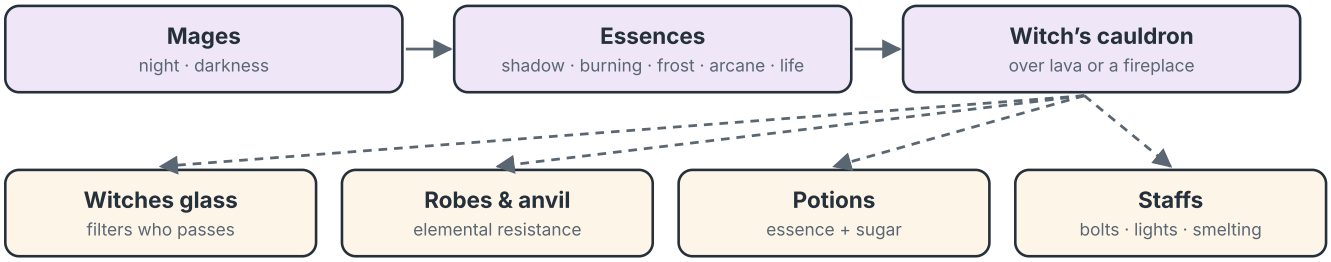
Machines, power and signals

The machine tree was built in eight phases (v0.18–v0.19) and extended through v0.43 (see Chapter 4 for worked wiring examples and the tech recipes). The map below is a rough guide to how the branches build on each other.



Branch	Key pieces and how they behave
Foundry	The metalwork crucible heats with fuel and only melts above 90% heat, with a spout on a face, into a tank beside it. 1 ingot = 144 mB, ore = 288. Pour onto a plate stand: a stick becomes a rod mould, and a mould casts rods.
Tanks and XP	Four tank tiers that connect and balance; same-type tanks merge into one clear structure and keep their contents when broken. XP is 20 mB per point; XP drains pull in orbs, and a tap pours while signalled.
Power (RF)	Windmills (about 1 RF/t, 10,000 RF stored, kept when broken), capacitor banks (50,000 RF), coal generators and solar panels I–VIII. Touching tech blocks share energy. Power lines reach 24 blocks per span, lose 0.4% per block (capped at 60%) and carry 20, 60 or 160 RF/t by coil tier.
Signals and logic	Levers give a little RF and a signal. One craftable gate does AND, OR, NOT, XOR, NAND, NOR, XNOR, BUF, LATCH, TFF, CLOCK and PULSE, with one tick of delay each. Signal levels and three tiers of signal blocks came in v0.20.
Lamps	16 colours, light level 15 spread over 15 blocks, posts on the ground and brackets on walls and ceilings; they follow an adjacent signal.
Silicon	A hand crank turns a sifter (20 steps per sand) for coal, iron shards, silicon and more. Moulder: 2 silicon → disc. Silicon press: disc + 2 fine copper wire → basic chip. Greater circuit press: advanced chip.
Terminals and wireless	RF terminals show a network's storage, generation, use and loss, and merge into bigger screens. Wireless broadcasters and receivers share channels (a linker copies them). Chunk loading keeps a 3×3 chunk area running.
Pistons	Push up to 12 blocks; the slimy piston pulls one back. They work in chunk-loaded areas.
Coloured wires (v0.38)	16 colours of wire laid on block faces (copper wire + wool). One wire carries both signals (the strongest wins) and RF (shared out by demand). Runs turn corners like rails; tap a second block in line and the run between is laid for you; tap a wired face with an empty hand to read the line.
Bundles and isolators (v0.38–v0.41)	Wire bundles carry all 16 colours at once. Bundle isolators are insulated conduits along one axis (X, Y or Z) that only connect at their ends, so rows of them run side by side without cross-talk.
Light shows and displays (v0.40–v0.41)	Multicolour lights pick one of 16 colours from a signal level, or mix RGB from three lines. A pixel driver turns a plane of lights into a 32×32 picture. Colour-locked logic blocks and isolators build multiplexed displays.
Logistics (v0.20)	Item distributors (pipes) with filters, hoppers and droppers, upgrades for chests and backpacks, placed backpacks.

Magic branch

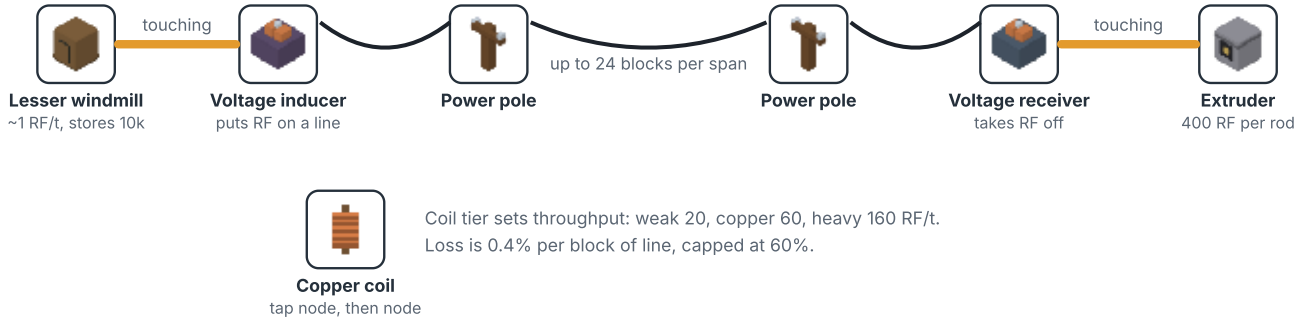


- **Mages** (illusion, fire, magick, ice, summoner) spawn at night or in darkness in Survival and drop shadow, burning, frost, arcane and life essences.
- **Fireplaces** burn forever; throw dye in to colour the smoke. The **witch's cauldron** heats over lava or a fireplace.
- **Witches glass** (essence + sand in the cauldron): shadow lets only the player through, burning only mobs, life only passive mobs and the player, frost only loose items.
- **Robes and the anvil**: brew leather armour with an essence for resistance, then line any armour with it. **Potions** come from bottled essence plus sugar. **Staffs** (stick + 50 essence in an activated cauldron) shoot bolts, place mage lights and smelt ore.

Wiring and machine demonstrations

Worked examples you can build, using the real blocks. Icons are the game's own item icons; every recipe is in the crafting guide.

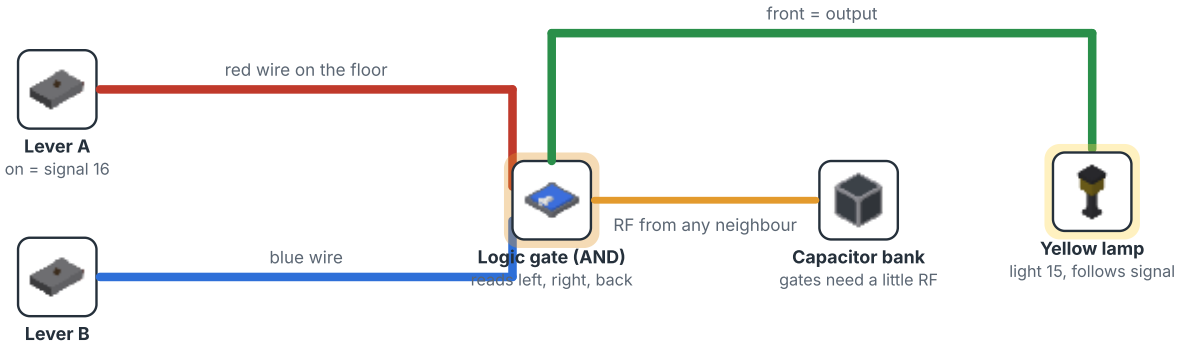
1. Power lines: from a windmill to a machine



Touching tech blocks share energy directly. For distance, tap a pole, inducer or receiver with a coil, then tap the next node. Each span uses one coil and draws as a sagging black wire.

1. Place the windmill high and in the open (it makes more with height and open sides), with a voltage inducer touching it.
2. Hold a coil, tap the inducer, then tap a power pole up to 24 blocks away. Chain poles as needed.
3. End at a voltage receiver touching the machine, or a capacitor bank (50,000 RF) to store the surplus. An RF terminal on the network shows storage, generation, use and loss.

2. Coloured wires, a logic gate and a lamp



A gate reads its left, right and back neighbours and outputs from its front, one tick later. Pick its type in its panel: AND, OR, NOT, XOR, NAND, NOR, XNOR, BUF, LATCH, TFF, CLOCK or PULSE.

Laying wire

- Craft coloured wire from copper wire and wool of that colour (4 at a time).
- Tap a block face to lay a segment. Tap a second block on the same face in a straight line and the run between is laid for you.
- Wires turn outside and inside corners like rails, and glow when live. Their braid crawls in the direction power flows, faster for a stronger signal.
- Tap a wired face with an empty hand to read the line. Breaking the block takes its wires with it.

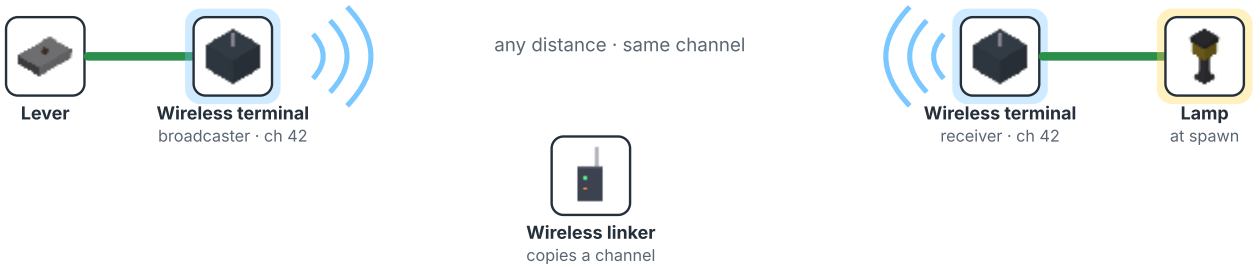
What a wire carries

- **Signals:** levels 0–16 that never fade. The strongest source on a line wins. Gates, lamps, taps and pistons treat anything above 0 as on.
- **RF:** every producer's output is shared out across every machine it can reach, by demand, through any colour.
- Different colours are separate signal lines, even side by side.
- Circuits update 20 times a second, and each gate or wireless hop takes exactly one tick.



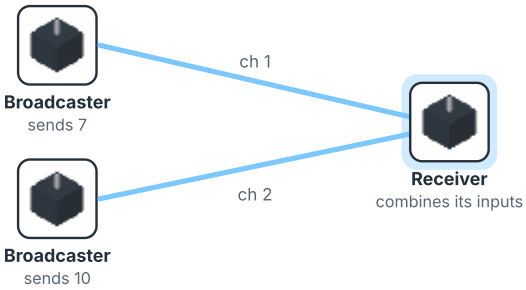
The Showcase PISTONS plot: a clock drives a slimy piston back and forth.

3. Wireless and signal levels



The Showcase WIRELESS plot: its lever lights a lamp at spawn on channel 42. Use the linker to copy a broadcaster's channel to a receiver.

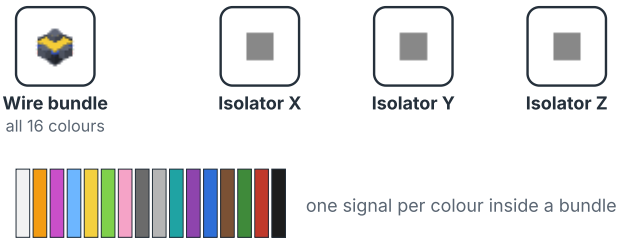
Block	Signals	Recipe in short
Signal broadcaster / receiver	Wired only, to the blocks touching them. Face roles (In, Out, Off), 14 operations, exact, preset or on/off output.	Basic logic chip, fine copper wire, stone, a coil
Wireless terminal (basic)	On/off over a channel. A broadcaster sends "on" while any input is on.	Radio control circuit, iron, glass, coils
Advanced wireless terminal	Levels and logic over channels: PASS, AND, OR, XOR, NAND, NOR, XNOR, ADD, SUB, MIN, MAX, EQ, GT, INV.	Steel table: steel, advanced chip, a basic terminal, a signal broadcaster and receiver



AND	7 AND 10 = **2**
OR	7 OR 10 = **15**
XOR	7 XOR 10 = **13**
ADD	7 ADD 10 = **16 (capped)**
SUB	7 SUB 10 = **3**
MIN	7 MIN 10 = **7**
NAND	7 NAND 10 = **14**

The Showcase SIGNAL LEVELS plot. These results were checked in the game: levels are 0–16, and the bitwise operations work on the numbers.

4. Bundles, isolators and light displays



Isolators only join at their two ends, never sideways, so 64 of them can lie shoulder to shoulder without cross-talk. A bundle over stone makes two; craft an isolator alone to turn its axis.



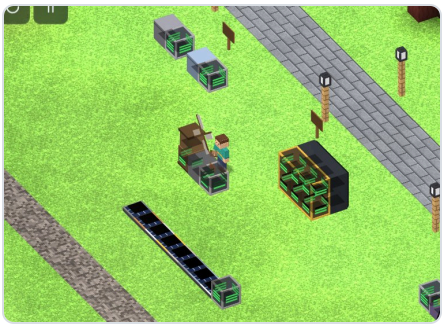
A multicolour light picks one of 16 colours from its signal level, or mixes red, green and blue lines (usually a bundle) in RGB mode. A pixel driver maps a wall of lights onto a 32×32 animation while powered.

5. In the Showcase world

Create a world with the **Showcase** type to walk through all of this already built, with a sign on every plot. The tech row is south of the main avenue; there's also a 64×40 upright display driven by a scan computer, and a gate lab of flip-flops and a counter made only of NAND gates.



POWER LINES: inducer → poles → receiver, with sagging lines.



POWER: windmill, a solar row, the capacitor multiblock and a terminal.



LOGIC GATES: two rows of gates with levers to flip.



FOUNDRY: a hot crucible, tank, spout and plate stand.



EXTRUDERS: rods into wire, plates, gears and tubes.



DISTRIBUTOR & HOPPER: pipes split at forks.

Strata features not in vanilla Minecraft 1.12.2

What you get in Strata's own worlds that vanilla 1.12.2 doesn't have. On a Minecraft server, the server's rules apply instead, so these are single-player and Strata-world features.

Area	In Strata	Vanilla 1.12.2
World	Worlds up to 10,000 × 10,000 with three depths (64, 128, 256 tall)	Infinite, 256 tall
	Woods: mangrove, cherry, pale oak, crimson, warped (11 in all)	6 woods
	Kelp, coral, sea lanterns without guardians	Kelp and coral came in 1.13
	Giant trees from four saplings in a square, for every wood	Only some woods
	Deepslate, tuff, calcite and their polished and brick forms	Not until 1.17
	The Abandoned Castle near spawn; a flat Showcase world	—
Dimensions	The Underworld (8:1, lava seas, flame towers, fire demons)	The Nether is different
	The Dusk: Lost villages, star-island libraries, the Telelink timedoor	—
	Pocket dimensions	—
Movement	Wall climbing with a stamina meter, dash, sprint toggle	—
	Hang glider; hookshots in every tier	Elytra only
	Elevators in 16 colours (jump up, crouch down)	—
	Iso camera with rotation and tilt, close follow when hidden, first person	First and third person
	1.13-style swimming	Came in 1.13
Survival	Eleven material tiers: tin, bronze, aluminium, steel, titanium, platinum and more, plus silver	Wood, stone, iron, gold, diamond
	Tiered crafting tables that gate recipes	One crafting table
	Smelters and alloys (bronze, steel), tanning table, woodworking table	Furnace only
	Hammers, knives, glaives and heavy weapons in every tier, and a steel katana	Swords and axes
	Warp bow with torch, fire, ice, teleport and swap arrows	—
	Tiered chests, backpacks with upgrades, storage drawers	Shulker boxes only
	Fishing minigame, loot table and journal	Simple fishing
	Grimoires (enchanting) at a grimoire desk; books of experience	Enchanting table
Tech	RF power: windmills, coal generators, solar panels I–VIII, capacitor banks, power lines	—
	Foundry: crucible, spouts, plate stands, rods and moulds	—

Area	In Strata	Vanilla 1.12.2
	Tanks in four tiers, XP tanks, drains and taps	—
	Logic gates (12 types), signal levels 0–16, signal broadcasters and receivers	Redstone only
	Coloured wires on block faces, bundles and isolators, sharing signal and RF	Redstone dust
	Wireless terminals with channels and level logic	—
	16-colour lamps, multicolour lights, pixel drivers and displays	Redstone lamp
	Extruders, magnets, silicon, chips and circuit presses	—
	Item distributors with filters, low-profile hoppers	Hoppers
	Chunk loading on any machine	Spawn chunks only
Magic	Five kinds of mage with elemental essences	—
	Witch's cauldron brewing; witches glass that filters who passes	—
	Essence robes and armour lining at the anvil	—
	Essence potions with sugar to extend them	Brewing stand
	Arcane, flaming, illusioner and frozen staffs	—
Creatures	Duskwalkers, fire demons, lava slimes, coloured slimes	—
	Squid and glow squid that ink and blind when hurt	Glow squid came in 1.17
	Every animal breeds, with heart particles	Most do
Creative	Paintings with a pixel editor, posters, camera photos	Fixed paintings
	ForgeWrites (rich-text books); signs with frames and backgrounds	Plain books and signs
	Layered banners with pattern, icon and emoji editor	Loom came in 1.14
	Instruments, MIDI import, custom music discs	Note blocks
	Dyeable campfires and fireplaces with coloured smoke	Campfire came in 1.14
	Beds and sleeping bags; weaving wheel; furniture tables and chairs	Beds only
Engine	Coloured block light that mixes through tinted glass	White light only
	Runs on a phone in one HTML file, touch-first	Java desktop

"—" means vanilla has nothing like it. Later Minecraft versions added their own takes on a few of these (noted where relevant).

Crafting guide: every recipe

All 1017 crafting recipes in Strata, read straight from the game, grouped as in the inventory's recipe list, plus every smelting recipe. Each card shows the grid, what it makes, and where you can craft it.

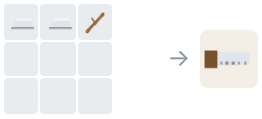
- **No table**: craftable in your inventory's grid.
- **Stone … Diamond crafting table**: you need that table, or a higher tier, within a few blocks.
- **Tanning / Woodworking table**: needs that station nearby.
- A small **+** on an ingredient means similar items work too (any planks, any log, any stone, and so on).
- The number on the result is how many you get.
- Tools, weapons and armour repeat for every material; their tier decides the table.

Smelting and alloys

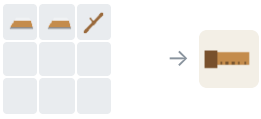
Input	Second input	Makes	Furnace (or better)	Time
Log + 35 similar		Charcoal	Lesser stone crucible	8 s
Sand + 1 similar		Glass	Lesser stone crucible	8 s
Cobblestone		Stone	Lesser stone crucible	6 s
Hellsand		Fire glass	Lesser stone crucible	8 s
Fire glass + 16 similar	Demon eye	Dusk block	Bronze smelter	20 s
Fire stone		Fire chalk	Lesser stone crucible	4 s
Raw cod		Cooked cod	Lesser stone crucible	5 s
Raw salmon		Cooked salmon	Lesser stone crucible	5 s
Lava bucket		Obsidian	Bronze smelter	20 s
Silver ore		Silver ingot	Lesser stone crucible	12 s
Cactus		Green dye	Lesser stone crucible	6 s
Kelp		Plant fiber	Lesser stone crucible	3 s
Clay		Brick	Lesser stone crucible	10 s
Glass	Red dye	Dyed version	Lesser stone crucible	4 s
Tin ore		Tin ingot	Lesser stone crucible	10 s
Copper ore		Copper ingot	Lesser stone crucible	10 s
Gold ore		Gold ingot	Lesser stone crucible	12 s
Raw beef		Cooked beef	Lesser stone crucible	6 s

Input	Second input	Makes	Furnace (or better)	Time
Raw pork		Cooked pork	Lesser stone crucible	6 s
Raw chicken		Cooked chicken	Lesser stone crucible	5 s
Copper ingot	Tin ingot	Bronze ingot ×2	Bronze smelter	12 s
Bauxite ore		Aluminium ingot	Bronze smelter	14 s
Iron ore		Iron ingot	Bronze smelter	14 s
Iron ingot	Coal	Steel ingot	Steel forge	18 s
Titanium ore		Titanium ingot	Steel forge	20 s
Platinum ore		Platinum ingot	Steel forge	22 s
Stone		Smooth stone	Lesser stone crucible	6 s
Potato		Baked potato	Lesser stone crucible	5 s
Kelp		Dried kelp	Lesser stone crucible	4 s
Cobbled deepslate		Deepslate	Lesser stone crucible	6 s
Cactus		Green dye	Lesser stone crucible	5 s
Iron shard		Iron nugget	Lesser stone crucible	4 s

Tools (104 recipes)



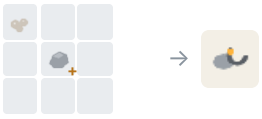
Aluminium hand saw
No table



Bronze hand saw
No table



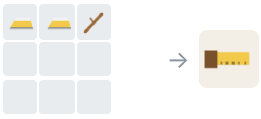
Copper hand saw
No table



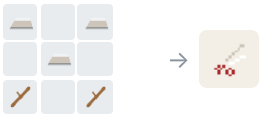
Fire starter
No table



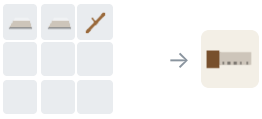
Fishing rod
No table



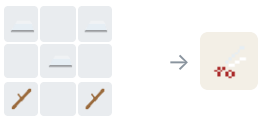
Gold hand saw
No table



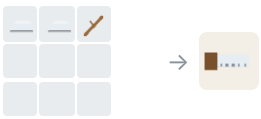
Iron cutters
No table



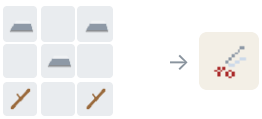
Iron hand saw
No table



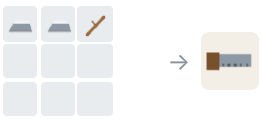
Platinum cutters
No table



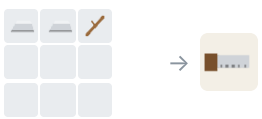
Platinum hand saw
No table



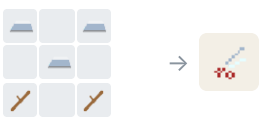
Steel cutters
No table



Steel hand saw
No table



Tin hand saw
No table



Titanium cutters
No table



Titanium hand saw
No table



Weak axe
No table



Weak bow
No table



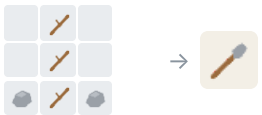
Weak hammer
No table



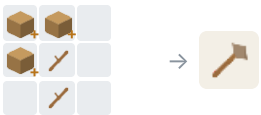
Weak knife
No table



Weak pickaxe
No table



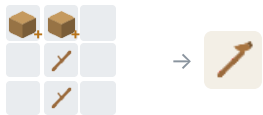
Weak shovel
No table



Wooden axe
No table



Wooden hammer
No table



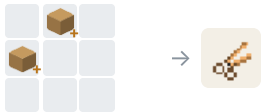
Wooden hoe
No table



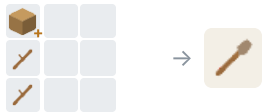
Wooden hookshot
No table



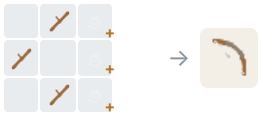
Wooden pickaxe
No table



Wooden shears
No table



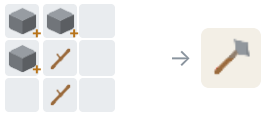
Wooden shovel
No table



Bow
Stone crafting table



Stone axe
Stone crafting table



Stone axe
Stone crafting table



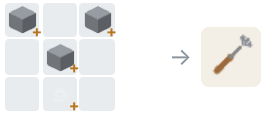
Stone hammer
Stone crafting table



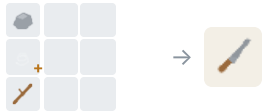
Stone hammer
Stone crafting table



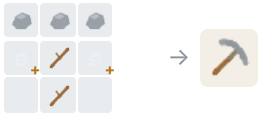
Stone hoe
Stone crafting table



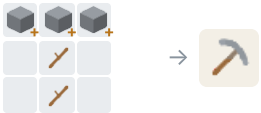
Stone hookshot
Stone crafting table



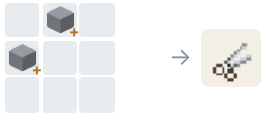
Stone knife
Stone crafting table



Stone pickaxe
Stone crafting table



Stone pickaxe
Stone crafting table



Stone shears
Stone crafting table



Stone shovel
Stone crafting table



Stone shovel
Stone crafting table



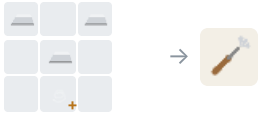
Tin axe
Tin crafting table



Tin hammer
Tin crafting table



Tin hoe
Tin crafting table



Tin hookshot
Tin crafting table



Tin pickaxe
Tin crafting table



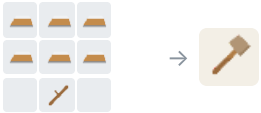
Tin shears
Tin crafting table



Tin shovel
Tin crafting table



Bronze axe
Bronze crafting table



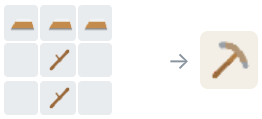
Bronze hammer
Bronze crafting table



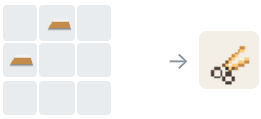
Bronze hoe
Bronze crafting table



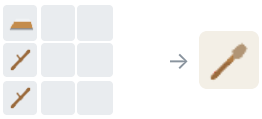
Bronze hookshot
Bronze crafting table



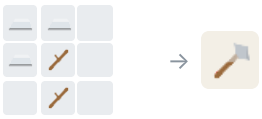
Bronze pickaxe
Bronze crafting table



Bronze shears
Bronze crafting table



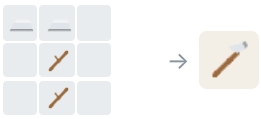
Bronze shovel
Bronze crafting table



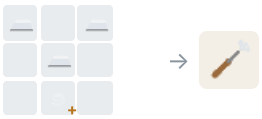
Aluminium axe
Aluminium crafting table



Aluminium hammer
Aluminium crafting table



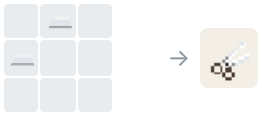
Aluminium hoe
Aluminium crafting table



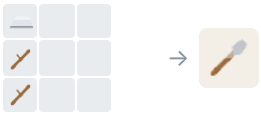
Aluminium hookshot
Aluminium crafting table



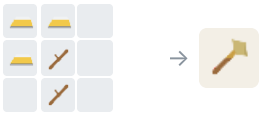
Aluminium pickaxe
Aluminium crafting table



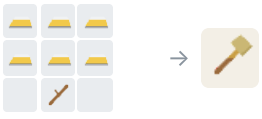
Aluminium shears
Aluminium crafting table



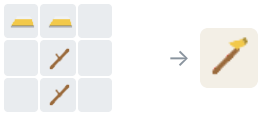
Aluminium shovel
Aluminium crafting table



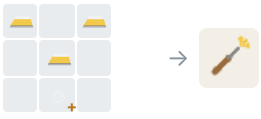
Gold axe
Gold crafting table



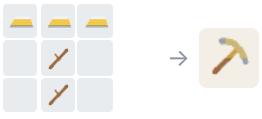
Gold hammer
Gold crafting table



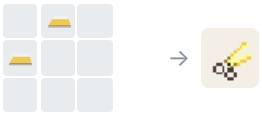
Gold hoe
Gold crafting table



Gold hookshot
Gold crafting table



Gold pickaxe
Gold crafting table



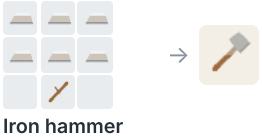
Gold shears
Gold crafting table



Gold shovel
Gold crafting table



Iron axe
Iron crafting table



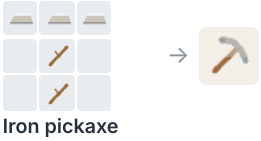
Iron hammer
Iron crafting table



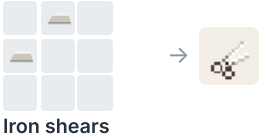
Iron hoe
Iron crafting table



Iron hookshot
Iron crafting table



Iron pickaxe
Iron crafting table



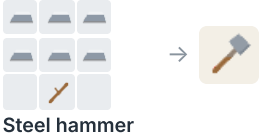
Iron shears
Iron crafting table



Iron shovel
Iron crafting table



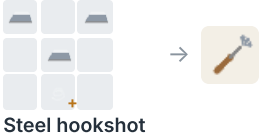
Steel axe
Steel crafting table



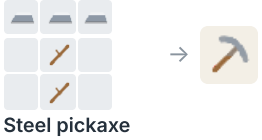
Steel hammer
Steel crafting table



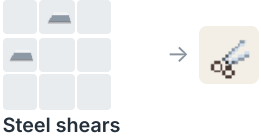
Steel hoe
Steel crafting table



Steel hookshot
Steel crafting table



Steel pickaxe
Steel crafting table



Steel shears
Steel crafting table



Steel shovel
Steel crafting table



Titanium axe
Titanium crafting table



Titanium hammer
Titanium crafting table



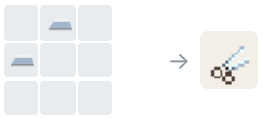
Titanium hoe
Titanium crafting table



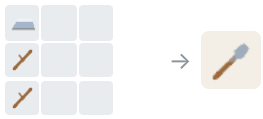
Titanium hookshot
Titanium crafting table



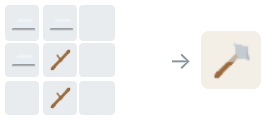
Titanium pickaxe
Titanium crafting table



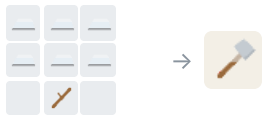
Titanium shears
Titanium crafting table



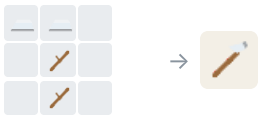
Titanium shovel
Titanium crafting table



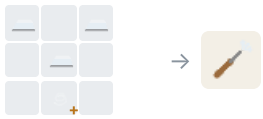
Platinum axe
Platinum crafting table



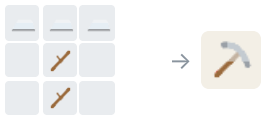
Platinum hammer
Platinum crafting table



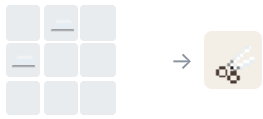
Platinum hoe
Platinum crafting table



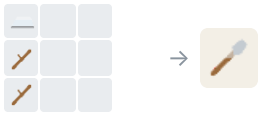
Platinum hookshot
Platinum crafting table



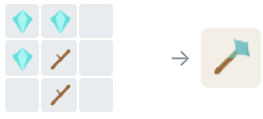
Platinum pickaxe
Platinum crafting table



Platinum shears
Platinum crafting table



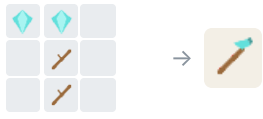
Platinum shovel
Platinum crafting table



Diamond axe
Diamond crafting table



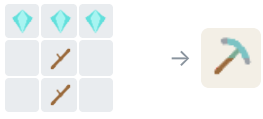
Diamond hammer
Diamond crafting table



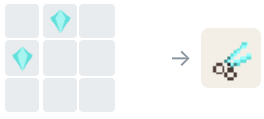
Diamond hoe
Diamond crafting table



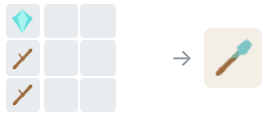
Diamond hookshot
Diamond crafting table



Diamond pickaxe
Diamond crafting table

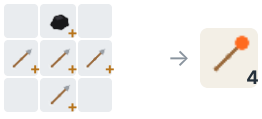


Diamond shears
Diamond crafting table

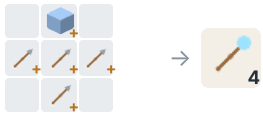


Diamond shovel
Diamond crafting table

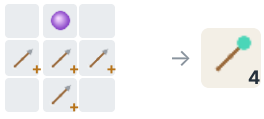
Weapons (87 recipes)



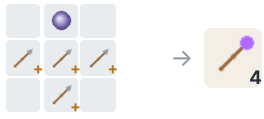
Fire arrow
No table



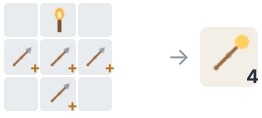
Ice arrow
No table



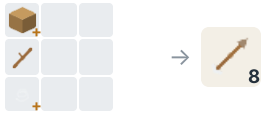
Swap arrow
No table



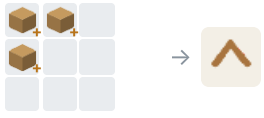
Teleport arrow
No table



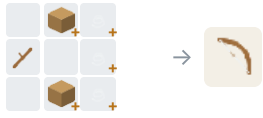
Torch arrow
No table



Wooden arrow
No table



Wooden boomerang
No table



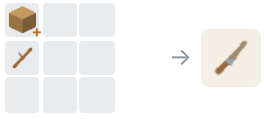
Wooden bow
No table



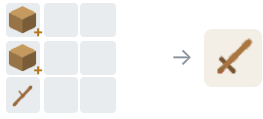
Wooden glaive
No table



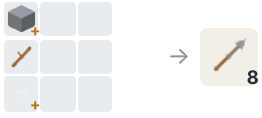
Wooden heavy axe
No table



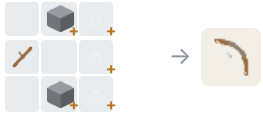
Wooden knife
No table



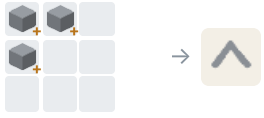
Wooden sword
No table



Arrow
Stone crafting table



Bow
Stone crafting table



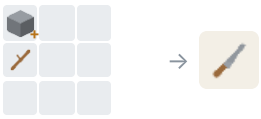
Stone boomerang
Stone crafting table



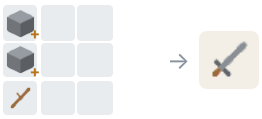
Stone glaive
Stone crafting table



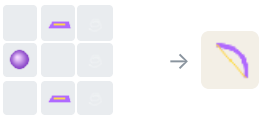
Stone heavy axe
Stone crafting table



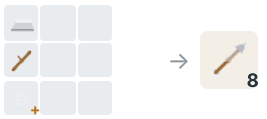
Stone knife
Stone crafting table



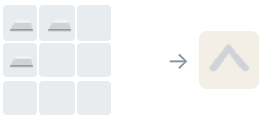
Stone sword
Stone crafting table



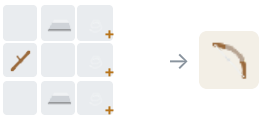
Warp bow
Stone crafting table



Tin arrow
Tin crafting table



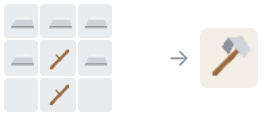
Tin boomerang
Tin crafting table



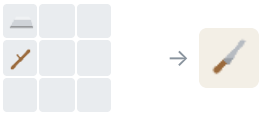
Tin bow
Tin crafting table



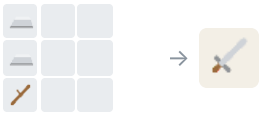
Tin glaive
Tin crafting table



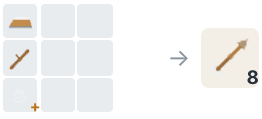
Tin heavy axe
Tin crafting table



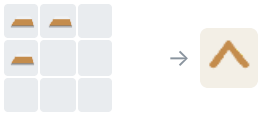
Tin knife
Tin crafting table



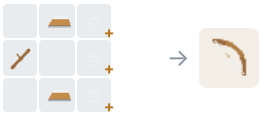
Tin sword
Tin crafting table



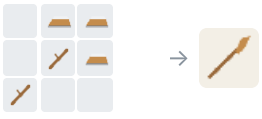
Bronze arrow
Bronze crafting table



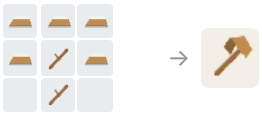
Bronze boomerang
Bronze crafting table



Bronze bow
Bronze crafting table



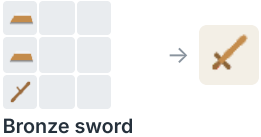
Bronze glaive
Bronze crafting table



Bronze heavy axe
Bronze crafting table



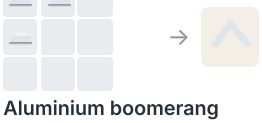
Bronze knife
Bronze crafting table



Bronze sword
Bronze crafting table



Aluminium arrow
Aluminium crafting table



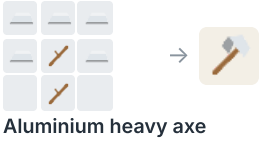
Aluminium boomerang
Aluminium crafting table



Aluminium bow
Aluminium crafting table



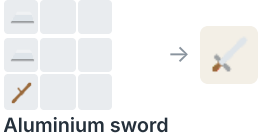
Aluminium glaive
Aluminium crafting table



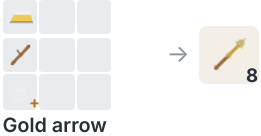
Aluminium heavy axe
Aluminium crafting table



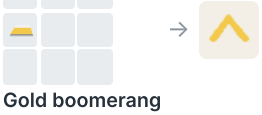
Aluminium knife
Aluminium crafting table



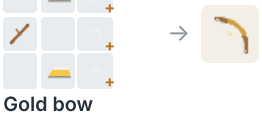
Aluminium sword
Aluminium crafting table



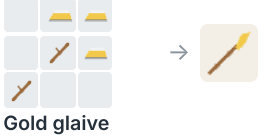
Gold arrow
Gold crafting table



Gold boomerang
Gold crafting table



Gold bow
Gold crafting table



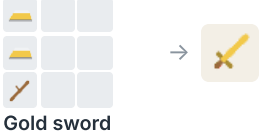
Gold glaive
Gold crafting table



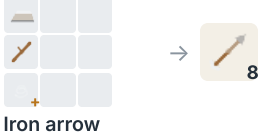
Gold heavy axe
Gold crafting table



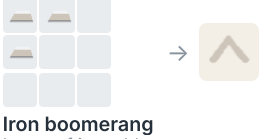
Gold knife
Gold crafting table



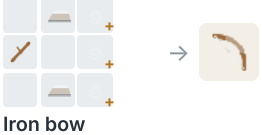
Gold sword
Gold crafting table



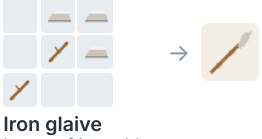
Iron arrow
Iron crafting table



Iron boomerang
Iron crafting table



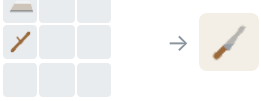
Iron bow
Iron crafting table



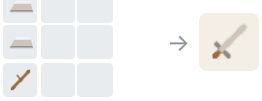
Iron glaive
Iron crafting table



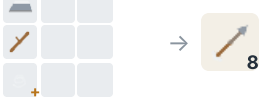
Iron heavy axe
Iron crafting table



Iron knife
Iron crafting table



Iron sword
Iron crafting table



Steel arrow
Steel crafting table



Steel boomerang
Steel crafting table



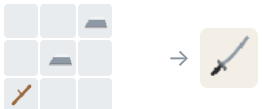
Steel bow
Steel crafting table



Steel glaive
Steel crafting table



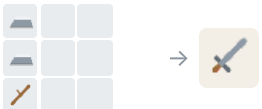
Steel heavy axe
Steel crafting table



Steel katana
Steel crafting table



Steel knife
Steel crafting table



Steel sword
Steel crafting table



Titanium arrow
Titanium crafting table



Titanium boomerang
Titanium crafting table



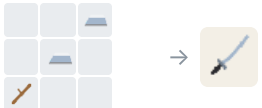
Titanium bow
Titanium crafting table



Titanium glaive
Titanium crafting table



Titanium heavy axe
Titanium crafting table



Titanium katana
Titanium crafting table



Titanium knife
Titanium crafting table



Titanium sword
Titanium crafting table



Platinum arrow
Platinum crafting table



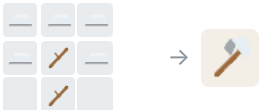
Platinum boomerang
Platinum crafting table



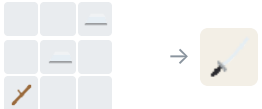
Platinum bow
Platinum crafting table



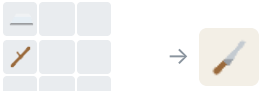
Platinum glaive
Platinum crafting table



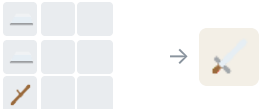
Platinum heavy axe
Platinum crafting table



Platinum katana
Platinum crafting table



Platinum knife
Platinum crafting table



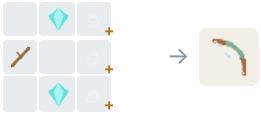
Platinum sword
Platinum crafting table



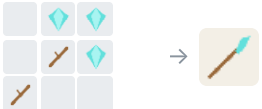
Diamond arrow
Diamond crafting table



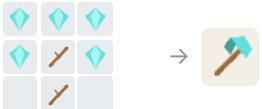
Diamond boomerang
Diamond crafting table



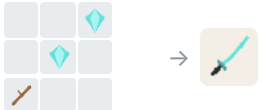
Diamond bow
Diamond crafting table



Diamond glaive
Diamond crafting table



Diamond heavy axe
Diamond crafting table



Diamond katana
Diamond crafting table

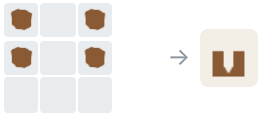


Diamond knife
Diamond crafting table

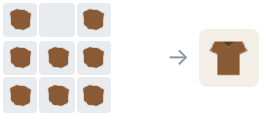


Diamond sword
Diamond crafting table

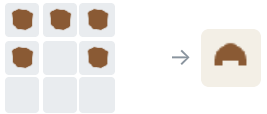
Armour (52 recipes)



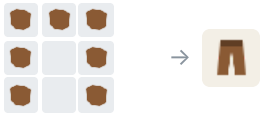
Leather boots
Tanning table



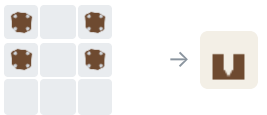
Leather chestplate
Tanning table



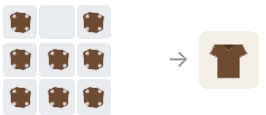
Leather helmet
Tanning table



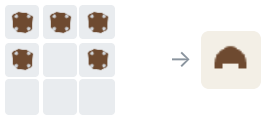
Leather leggings
Tanning table



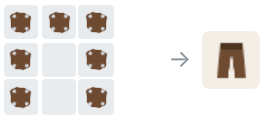
Reinforced leather boots
Tanning table



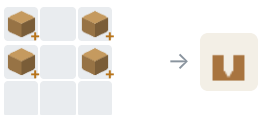
Reinforced leather chestplate
Tanning table



Reinforced leather helmet
Tanning table



Reinforced leather leggings
Tanning table



Wooden boots
No table



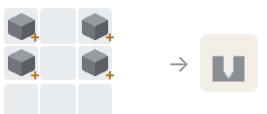
Wooden chestplate
No table



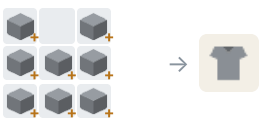
Wooden helmet
No table



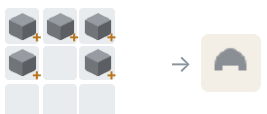
Wooden leggings
No table



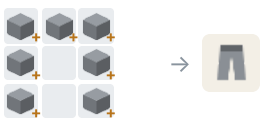
Stone boots
Stone crafting table



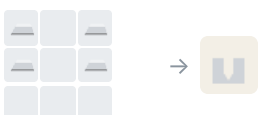
Stone chestplate
Stone crafting table



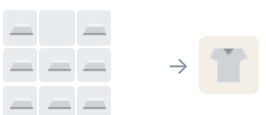
Stone helmet
Stone crafting table



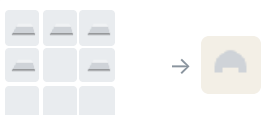
Stone leggings
Stone crafting table



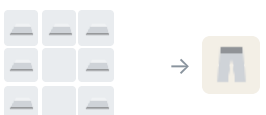
Tin boots
Tin crafting table



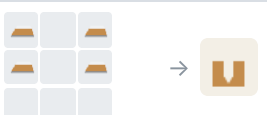
Tin chestplate
Tin crafting table



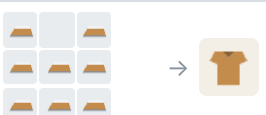
Tin helmet
Tin crafting table



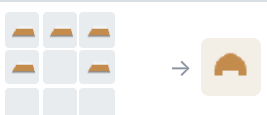
Tin leggings
Tin crafting table



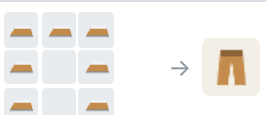
Bronze boots
Bronze crafting table



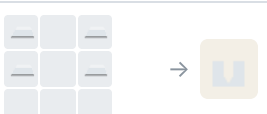
Bronze chestplate
Bronze crafting table



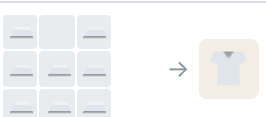
Bronze helmet
Bronze crafting table



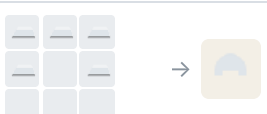
Bronze leggings
Bronze crafting table



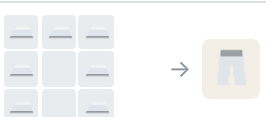
Aluminium boots
Aluminium crafting table



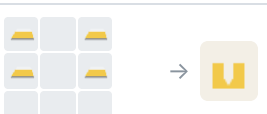
Aluminium chestplate
Aluminium crafting table



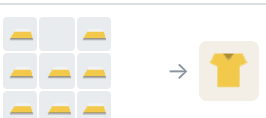
Aluminium helmet
Aluminium crafting table



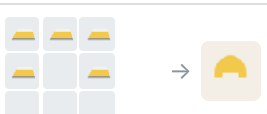
Aluminium leggings
Aluminium crafting table



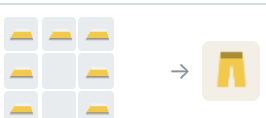
Gold boots
Gold crafting table



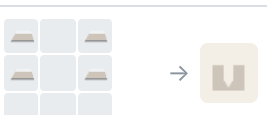
Gold chestplate
Gold crafting table



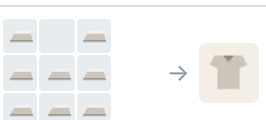
Gold helmet
Gold crafting table



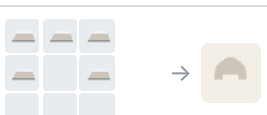
Gold leggings
Gold crafting table



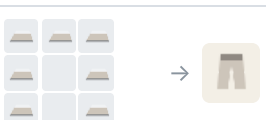
Iron boots
Iron crafting table



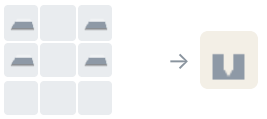
Iron chestplate
Iron crafting table



Iron helmet
Iron crafting table



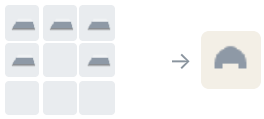
Iron leggings
Iron crafting table



Steel boots
Steel crafting table



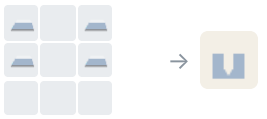
Steel chestplate
Steel crafting table



Steel helmet
Steel crafting table



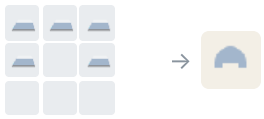
Steel leggings
Steel crafting table



Titanium boots
Titanium crafting table



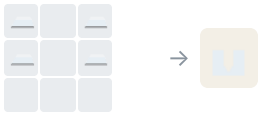
Titanium chestplate
Titanium crafting table



Titanium helmet
Titanium crafting table



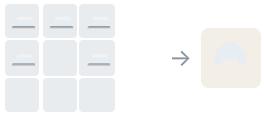
Titanium leggings
Titanium crafting table



Platinum boots
Platinum crafting table



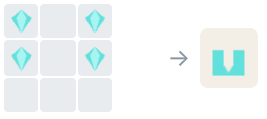
Platinum chestplate
Platinum crafting table



Platinum helmet
Platinum crafting table



Platinum leggings
Platinum crafting table



Diamond boots
Diamond crafting table



Diamond chestplate
Diamond crafting table

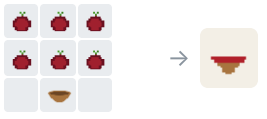


Diamond helmet
Diamond crafting table

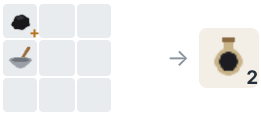


Diamond leggings
Diamond crafting table

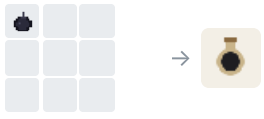
Items, food and materials (78 recipes)



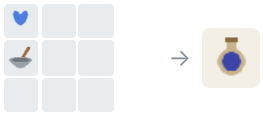
Beetroot soup
No table



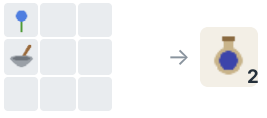
Black dye
No table



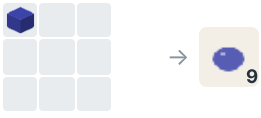
Black dye
No table



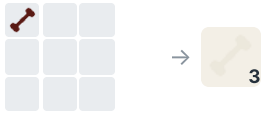
Blue dye
No table



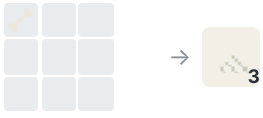
Blue dye
No table



Blue slime jelly
No table



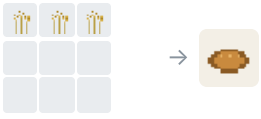
Bone
No table



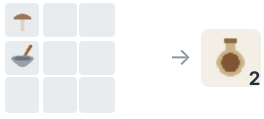
Bone meal
No table



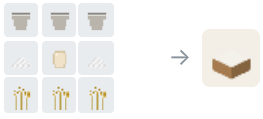
Book
No table



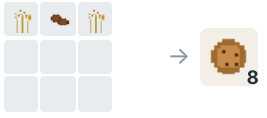
Bread
No table



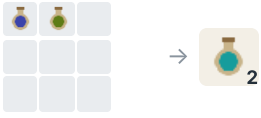
Brown dye
No table



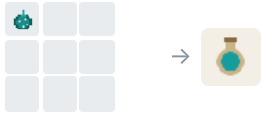
Cake
No table



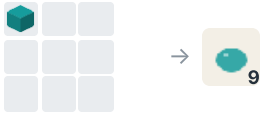
Cookie
No table



Cyan dye
No table



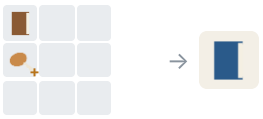
Cyan dye
No table



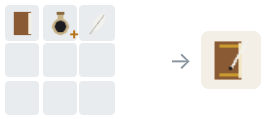
Cyan slime jelly
No table



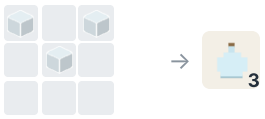
Empty map
No table



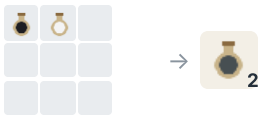
Fishing journal
No table



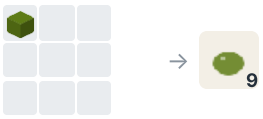
ForgeWrite
No table



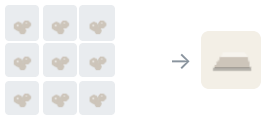
Glass bottle
No table



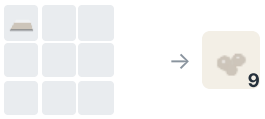
Gray dye
No table



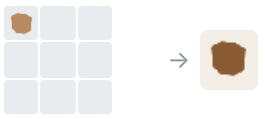
Green slime jelly
No table



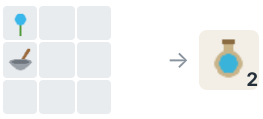
Iron ingot
No table



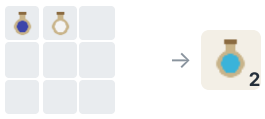
Iron nugget
No table



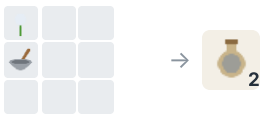
Leather
Tanning table



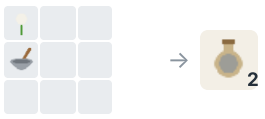
Light blue dye
No table



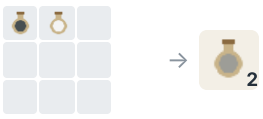
Light blue dye
No table



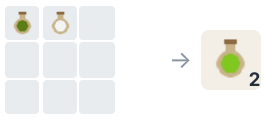
Light gray dye
No table



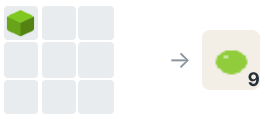
Light gray dye
No table



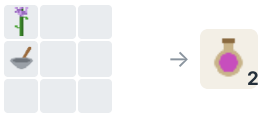
Light gray dye
No table



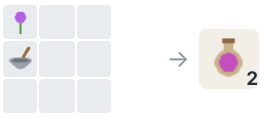
Lime dye
No table



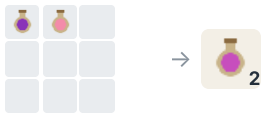
Lime slime jelly
No table



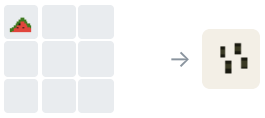
Magenta dye
No table



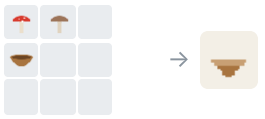
Magenta dye
No table



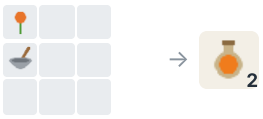
Magenta dye
No table



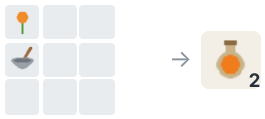
Melon seeds
No table



Mushroom stew
No table



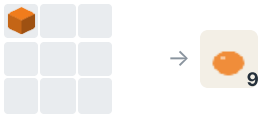
Orange dye
No table



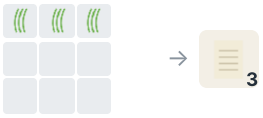
Orange dye
No table



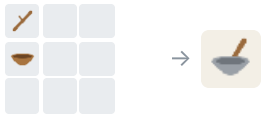
Orange dye
No table



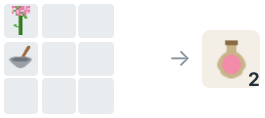
Orange slime jelly
No table



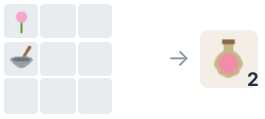
Paper
No table



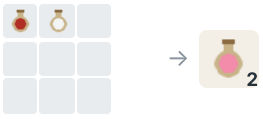
Pestle and mortar
No table



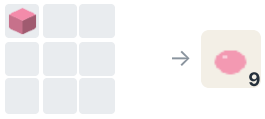
Pink dye
No table



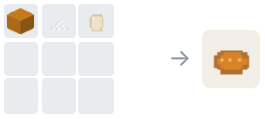
Pink dye
No table



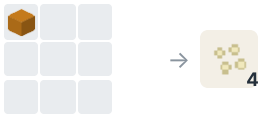
Pink dye
No table



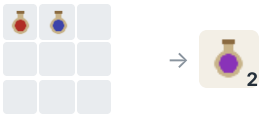
Pink slime jelly
No table



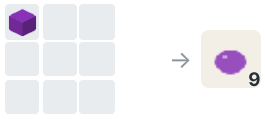
Pumpkin pie
No table



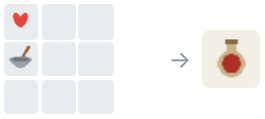
Pumpkin seeds
No table



Purple dye
No table



Purple slime jelly
No table



Red dye
No table



Red dye
No table



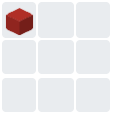
Red dye
No table



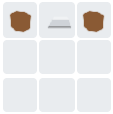
Red dye
No table



Red dye
No table



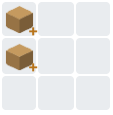
Red slime jelly
No table



Reinforced leather
Tanning table

Silver ingot
No table

Silver nugget
No table



Stick
No table



Sugar
No table



Weak arrow
No table



White dye
No table



White dye
No table



White dye
No table



White dye
No table

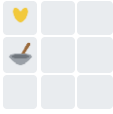
White dye
No table



Wood bowl
No table



Written ForgeWrite
No table



Yellow dye
No table



Yellow dye
No table



Yellow dye
No table



Arrow
Stone crafting table



Book of experience
Stone crafting table



Bucket
Stone crafting table



Camera
Stone crafting table

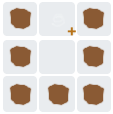


Warp ingot
Stone crafting table

Storage (20 recipes)



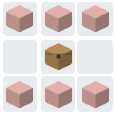
Acacia storage drawer
No table



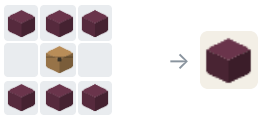
Backpack
Tanning table



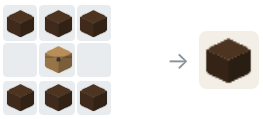
Birch storage drawer
No table



Cherry storage drawer
No table



Crimson storage drawer
No table



Dark oak storage drawer
No table



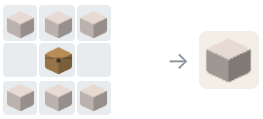
Jungle storage drawer
No table



Mangrove storage drawer
No table



Oak storage drawer
No table



Pale oak storage drawer
No table



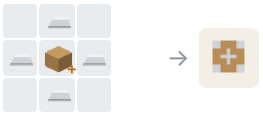
Spruce storage drawer
No table



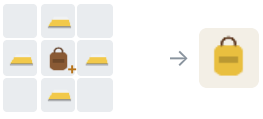
Warped storage drawer
No table



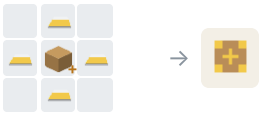
Wooden chest
No table



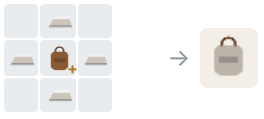
Tin chest upgrade
Tin crafting table



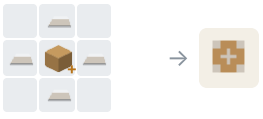
Gold backpack
Gold crafting table



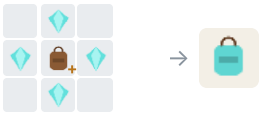
Gold chest upgrade
Gold crafting table



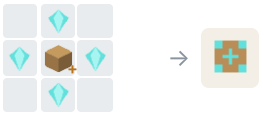
Iron backpack
Iron crafting table



Iron chest upgrade
Iron crafting table

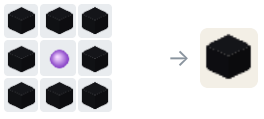


Diamond backpack
Diamond crafting table



Diamond chest upgrade
Diamond crafting table

Travel (32 recipes)



Black elevator
No table



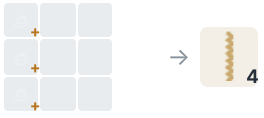
Blue elevator
No table



Boat
No table



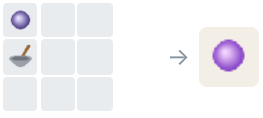
Brown elevator
No table



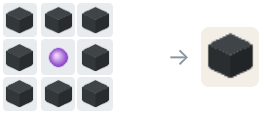
Climbing rope
No table



Cyan elevator
No table



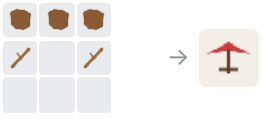
Dusk orb
No table



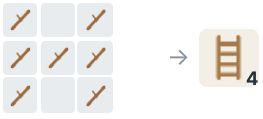
Gray elevator
No table



Green elevator
No table



Hang glider
No table



Ladder
No table



Light blue elevator
No table



Light gray elevator
No table



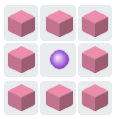
Lime elevator
No table



Magenta elevator
No table



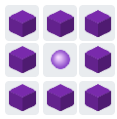
Orange elevator
No table



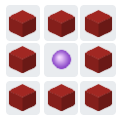
Pink elevator
No table



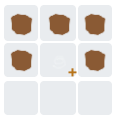
Pocket expander
No table



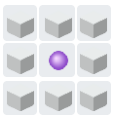
Purple elevator
No table



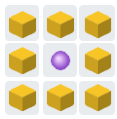
Red elevator
No table



Saddle
Tanning table



White elevator
No table



Yellow elevator
No table



Wooden minecart
No table



Lesser telelink
Stone crafting table



Pocket dimension
Stone crafting table



Rail
Tin crafting table



Bronze minecart
Bronze crafting table



Iron minecart
Iron crafting table



Steel minecart
Steel crafting table



Titanium minecart
Titanium crafting table



Platinum minecart
Platinum crafting table

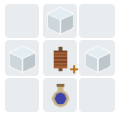
Tech (102 recipes)



Black bulb
No table



Black lamp
No table



Blue bulb
No table



Blue lamp
No table



Brown bulb
No table



Brown lamp
No table

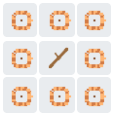


Capacitor bank
No table

Cloth
No table



Coal generator
No table



Copper coil
No table



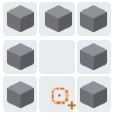
Crucible spout
No table



Cyan bulb
No table



Cyan lamp
No table



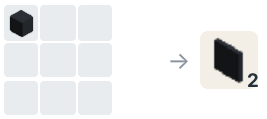
Dropper
No table



Extruder
No table



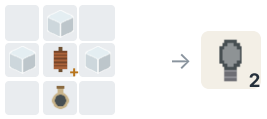
Filter upgrade
No table



Flat RF terminal
No table



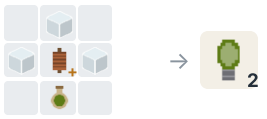
Glass tank
No table



Gray bulb
No table



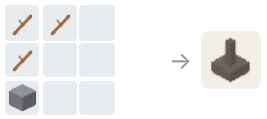
Gray lamp
No table



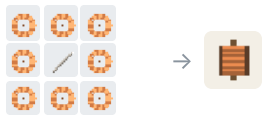
Green bulb
No table



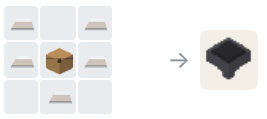
Green lamp
No table



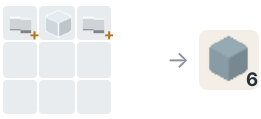
Hand crank
No table



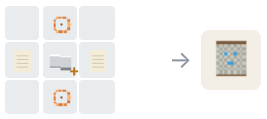
Heavy copper coil
No table



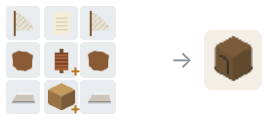
Hopper
No table



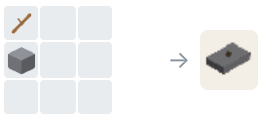
Item distributor
No table



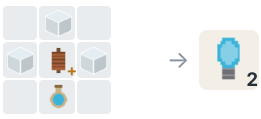
Item filter
No table



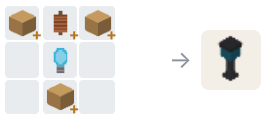
Lesser windmill
No table



Lever
No table



Light blue bulb
No table



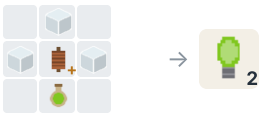
Light blue lamp
No table



Light gray bulb
No table



Light gray lamp
No table



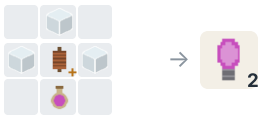
Lime bulb
No table



Lime lamp
No table



Logic gate
No table



Magenta bulb
No table



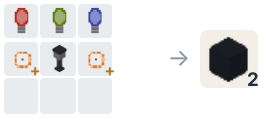
Magenta lamp
No table



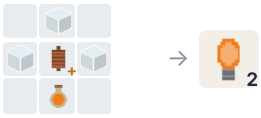
Metalwork crucible
No table



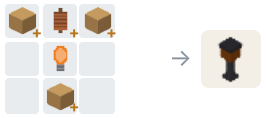
Mirror
No table



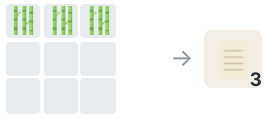
Multicolour light
No table



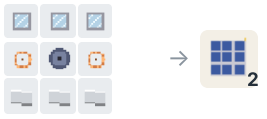
Orange bulb
No table



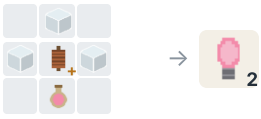
Orange lamp
No table



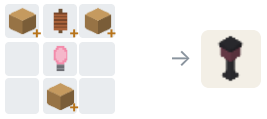
Paper
No table



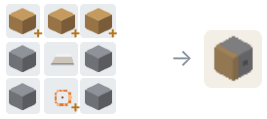
Photovoltaic cell I
No table



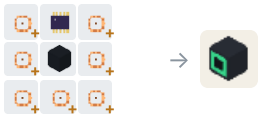
Pink bulb
No table



Pink lamp
No table



Piston
No table



Pixel driver
No table

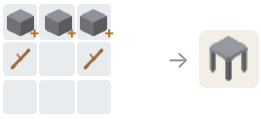
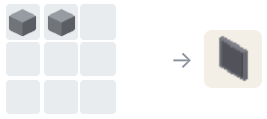


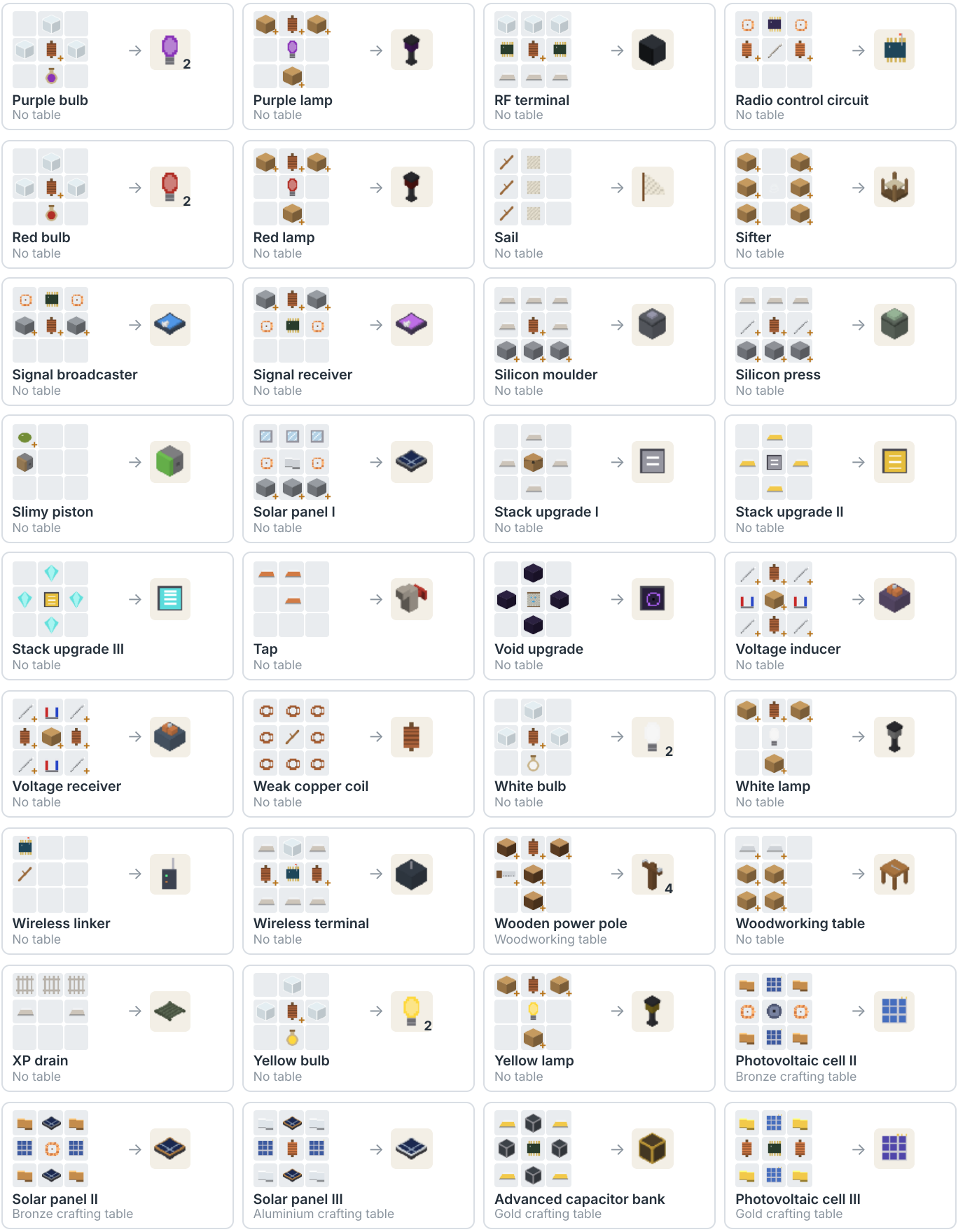
Plate stand
No table

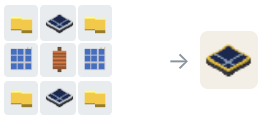


Poor copper wire
No table



Pressure plate (signal)
No table


Purple bulb
No table
Purple lamp
No table
RF terminal
No table
Radio control circuit
No table
Red bulb
No table
Red lamp
No table
Sail
No table
Sifter
No table
Signal broadcaster
No table
Signal receiver
No table
Silicon moulder
No table
Silicon press
No table
Slimy piston
No table
Solar panel I
No table
Stack upgrade I
No table
Stack upgrade II
No table
Stack upgrade III
No table
Tap
No table
Void upgrade
No table
Voltage inducer
No table
Voltage receiver
No table
Weak copper coil
No table
White bulb
No table
White lamp
No table
Wireless linker
No table
Wireless terminal
No table
Wooden power pole
Woodworking table
Woodworking table
No table
XP drain
No table
Yellow bulb
No table
Yellow lamp
No table
Photovoltaic cell II
Bronze crafting table
Solar panel II
Bronze crafting table
Solar panel III
Aluminium crafting table
Advanced capacitor bank
Gold crafting table
Photovoltaic cell III
Gold crafting table



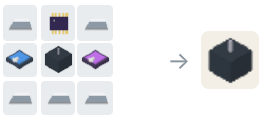
Solar panel IV
Gold crafting table



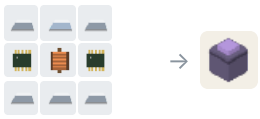
Reinforced tank
Iron crafting table



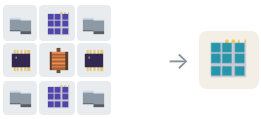
Solar panel V
Iron crafting table



Advanced wireless terminal
Steel crafting table



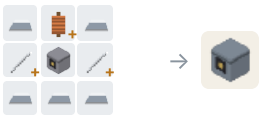
Greater circuit press
Steel crafting table



Photovoltaic cell IV
Steel crafting table



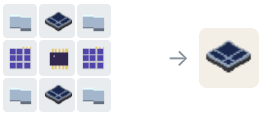
Solar panel VI
Steel crafting table



Steel extruder
Steel crafting table



Steel tank
Steel crafting table



Solar panel VII
Titanium crafting table



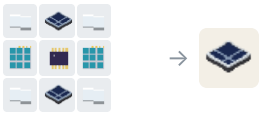
Titanium extruder
Titanium crafting table



Titanium tank
Titanium crafting table

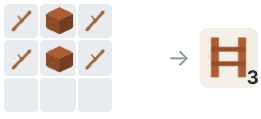


Vibrant capacitor bank
Titanium crafting table

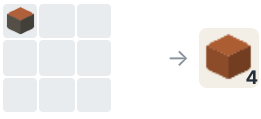


Solar panel VIII
Platinum crafting table

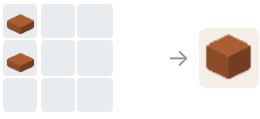
Blocks (323 recipes)



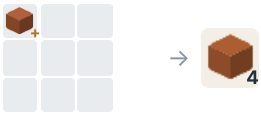
Acacia fence
No table



Acacia planks
No table



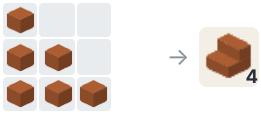
Acacia planks
No table



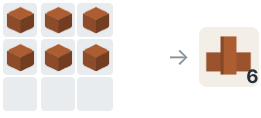
Acacia planks
No table



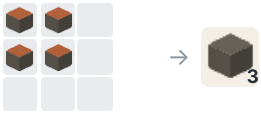
Acacia slab
No table



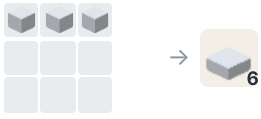
Acacia stairs
No table



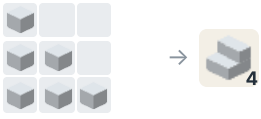
Acacia wall
No table



Acacia wood
No table



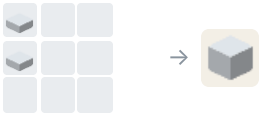
Aluminium brick slab
No table



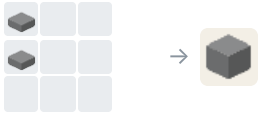
Aluminium brick stairs
No table



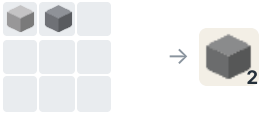
Aluminium brick wall
No table



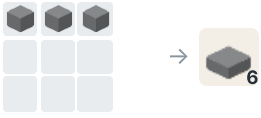
Aluminium bricks
No table



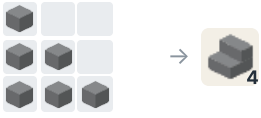
Andesite
No table



Andesite
No table



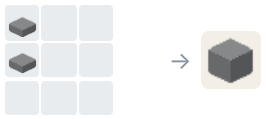
Andesite brick slab
No table



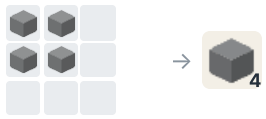
Andesite brick stairs
No table



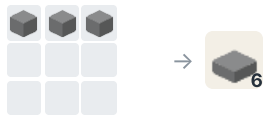
Andesite brick wall
No table



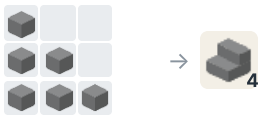
Andesite bricks
No table



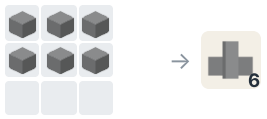
Andesite bricks
No table



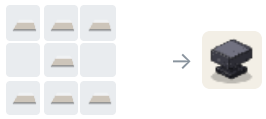
Andesite slab
No table



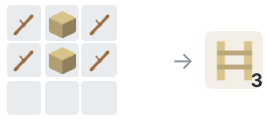
Andesite stairs
No table



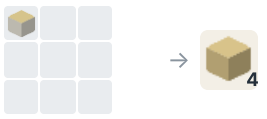
Andesite wall
No table



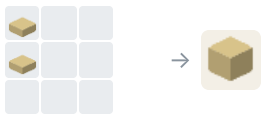
Anvil
No table



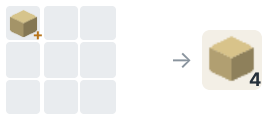
Birch fence
No table



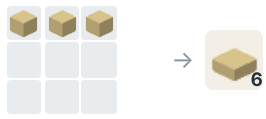
Birch planks
No table



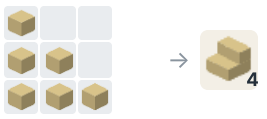
Birch planks
No table



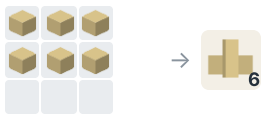
Birch planks
No table



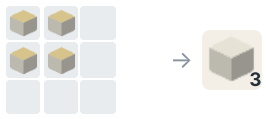
Birch slab
No table



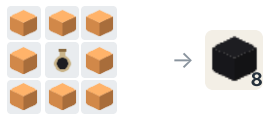
Birch stairs
No table



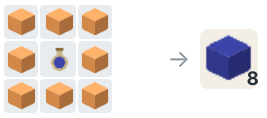
Birch wall
No table



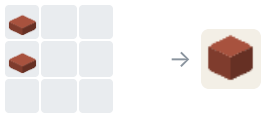
Birch wood
No table



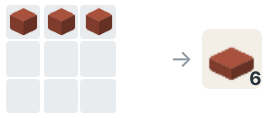
Black fire glass
No table



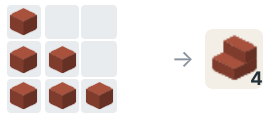
Blue fire glass
No table



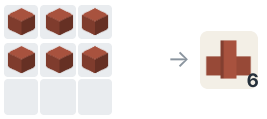
Brick
No table



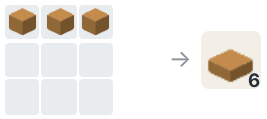
Brick slab
No table



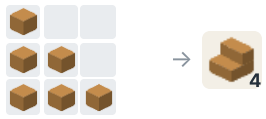
Brick stairs
No table



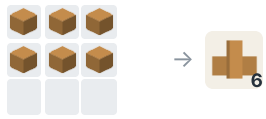
Brick wall
No table



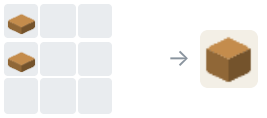
Bronze brick slab
No table



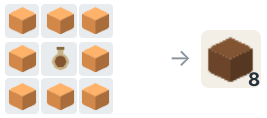
Bronze brick stairs
No table



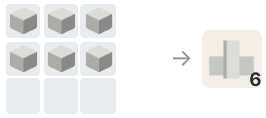
Bronze brick wall
No table



Bronze bricks
No table



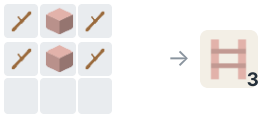
Brown fire glass
No table



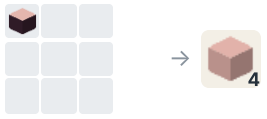
Calcite wall
No table



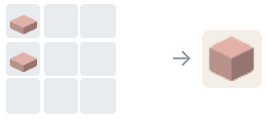
Campfire
No table



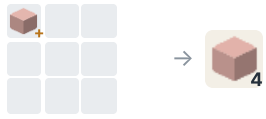
Cherry fence
No table



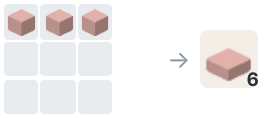
Cherry planks
No table



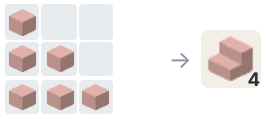
Cherry planks
No table



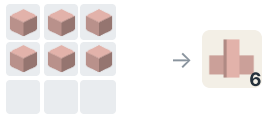
Cherry planks
No table



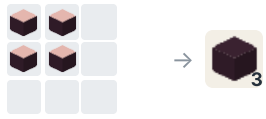
Cherry slab
No table



Cherry stairs
No table



Cherry wall
No table



Cherry wood
No table



Chiseled stone bricks
No table



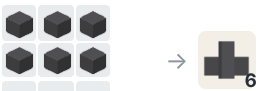
Cobbled deepslate
No table



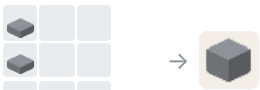
Cobbled deepslate slab
No table



Cobbled deepslate stairs
No table



Cobbled deepslate wall
No table



Cobblestone
No table



Cobblestone slab
No table



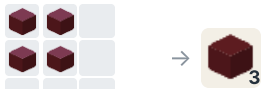
Cobblestone stairs
No table



Cobblestone wall
No table



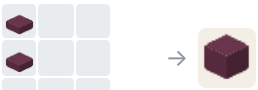
Crimson fence
No table



Crimson hyphae
No table



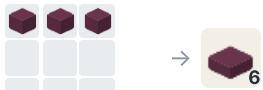
Crimson planks
No table



Crimson planks
No table



Crimson planks
No table



Crimson slab
No table



Crimson stairs
No table



Crimson wall
No table



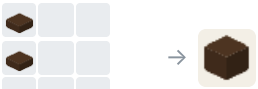
Cyan fire glass
No table



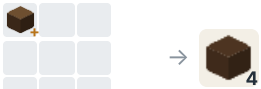
Dark oak fence
No table



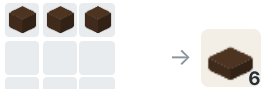
Dark oak planks
No table



Dark oak planks
No table



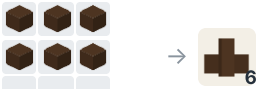
Dark oak planks
No table



Dark oak slab
No table



Dark oak stairs
No table



Dark oak wall
No table



Dark oak wood
No table



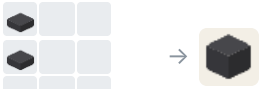
Deepslate brick slab
No table



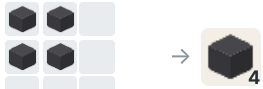
Deepslate brick stairs
No table



Deepslate brick wall
No table



Deepslate bricks
No table



Deepslate bricks
No table



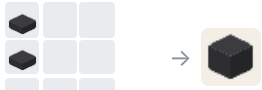
Deepslate tile slab
No table



Deepslate tile stairs
No table



Deepslate tile wall
No table



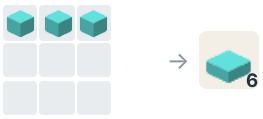
Deepslate tiles
No table



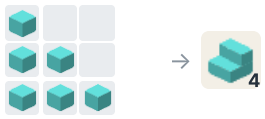
Deepslate tiles
No table



Deepslate wall
No table



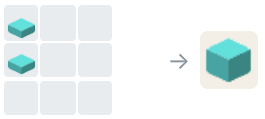
Diamond brick slab
No table



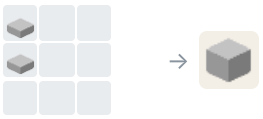
Diamond brick stairs
No table



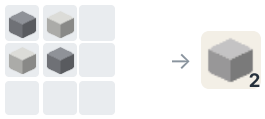
Diamond brick wall
No table



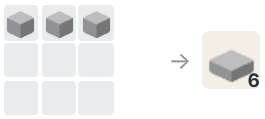
Diamond bricks
No table



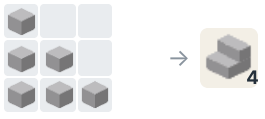
Diorite
No table



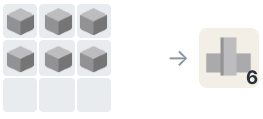
Diorite
No table



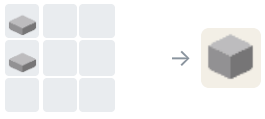
Diorite brick slab
No table



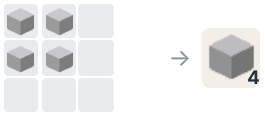
Diorite brick stairs
No table



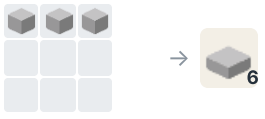
Diorite brick wall
No table



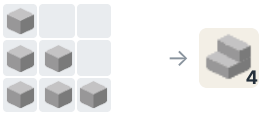
Diorite bricks
No table



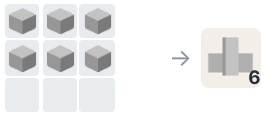
Diorite bricks
No table



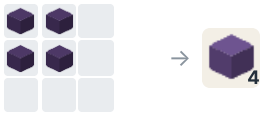
Diorite slab
No table



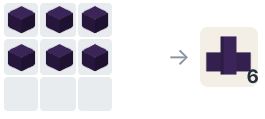
Diorite stairs
No table



Diorite wall
No table



Dusk bricks
No table



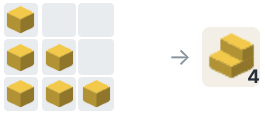
Dusk wall
No table



Fireplace
No table



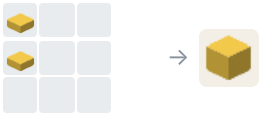
Gold brick slab
No table



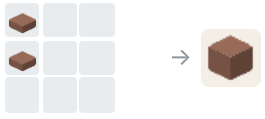
Gold brick stairs
No table



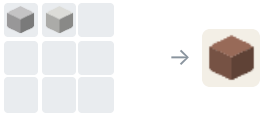
Gold brick wall
No table



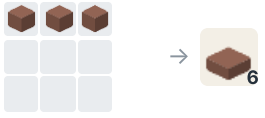
Gold bricks
No table



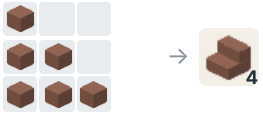
Granite
No table



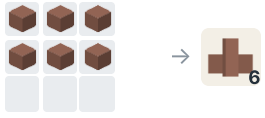
Granite
No table



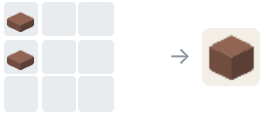
Granite brick slab
No table



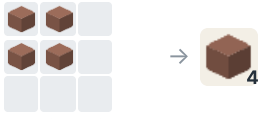
Granite brick stairs
No table



Granite brick wall
No table



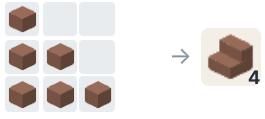
Granite bricks
No table



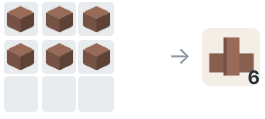
Granite bricks
No table



Granite slab
No table



Granite stairs
No table



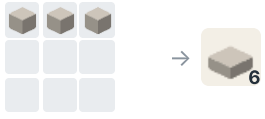
Granite wall
No table



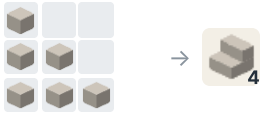
Gray fire glass
No table



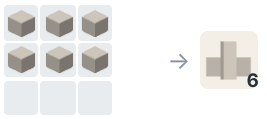
Green fire glass
No table



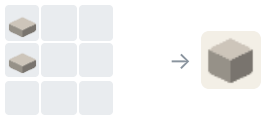
Iron brick slab
No table



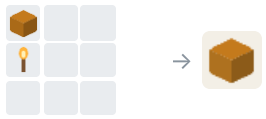
Iron brick stairs
No table



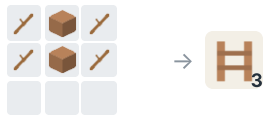
Iron brick wall
No table



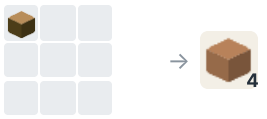
Iron bricks
No table



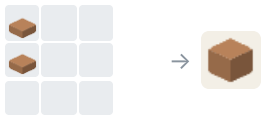
Jack o'lantern
No table



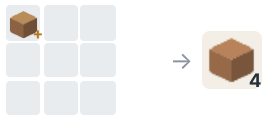
Jungle fence
No table



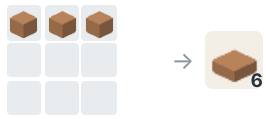
Jungle planks
No table



Jungle planks
No table



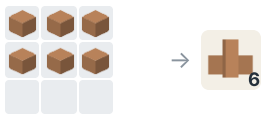
Jungle planks
No table



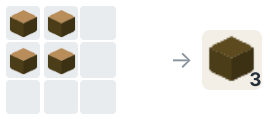
Jungle slab
No table



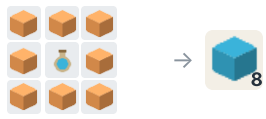
Jungle stairs
No table



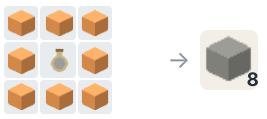
Jungle wall
No table



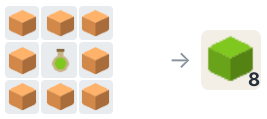
Jungle wood
No table



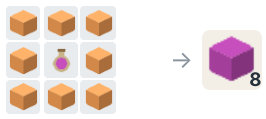
Light blue fire glass
No table



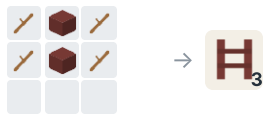
Light gray fire glass
No table



Lime fire glass
No table



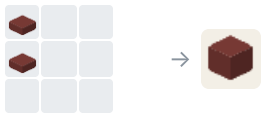
Magenta fire glass
No table



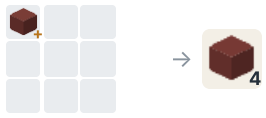
Mangrove fence
No table



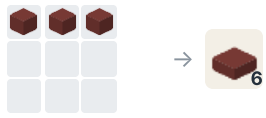
Mangrove planks
No table



Mangrove planks
No table



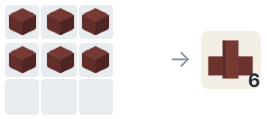
Mangrove planks
No table



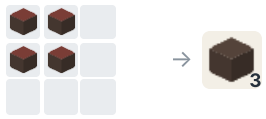
Mangrove slab
No table



Mangrove stairs
No table



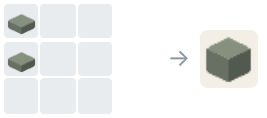
Mangrove wall
No table



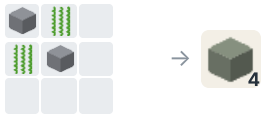
Mangrove wood
No table



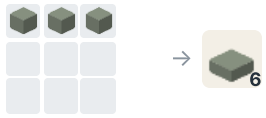
Melon
No table



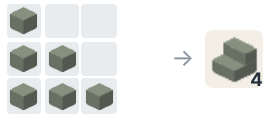
Mossy cobblestone
No table



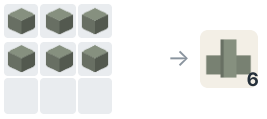
Mossy cobblestone
No table



Mossy cobblestone slab
No table



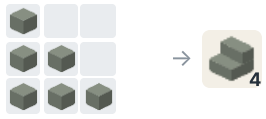
Mossy cobblestone stairs
No table



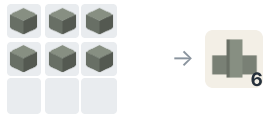
Mossy cobblestone wall
No table



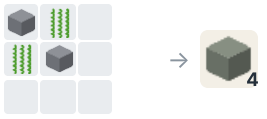
Mossy stone brick slab
No table



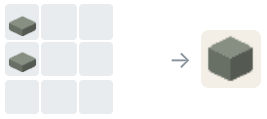
Mossy stone brick stairs
No table



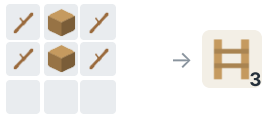
Mossy stone brick wall
No table



Mossy stone bricks
No table



Mossy stone bricks
No table



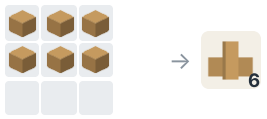
Oak fence
No table



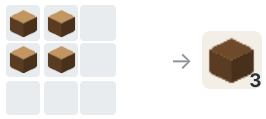
Oak slab
No table



Oak stairs
No table



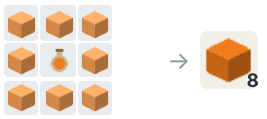
Oak wall
No table



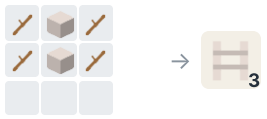
Oak wood
No table



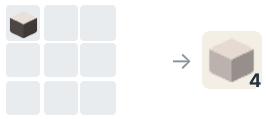
Obsidian wall
No table



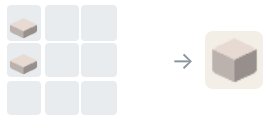
Orange fire glass
No table



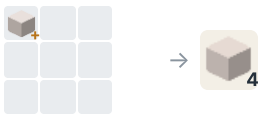
Pale oak fence
No table



Pale oak planks
No table



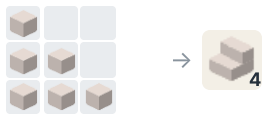
Pale oak planks
No table



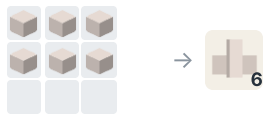
Pale oak planks
No table



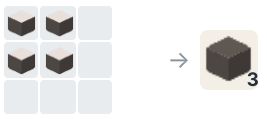
Pale oak slab
No table



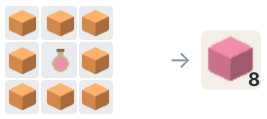
Pale oak stairs
No table



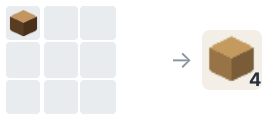
Pale oak wall
No table



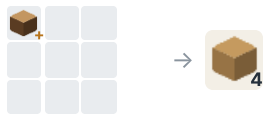
Pale oak wood
No table



Pink fire glass
No table



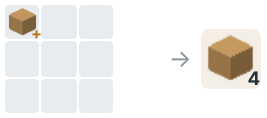
Planks
No table



Planks
No table



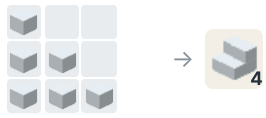
Planks
No table



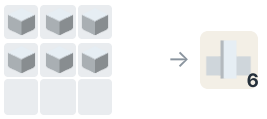
Planks
No table



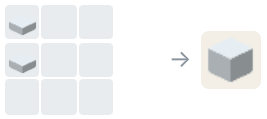
Platinum brick slab
No table



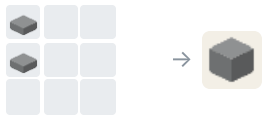
Platinum brick stairs
No table



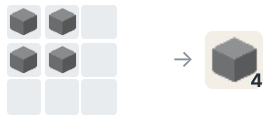
Platinum brick wall
No table



Platinum bricks
No table



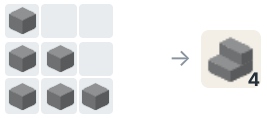
Polished andesite
No table



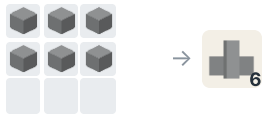
Polished andesite
No table



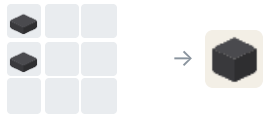
Polished andesite slab
No table



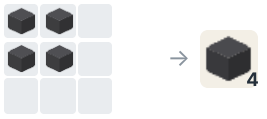
Polished andesite stairs
No table



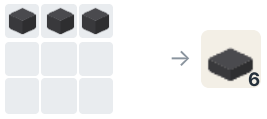
Polished andesite wall
No table



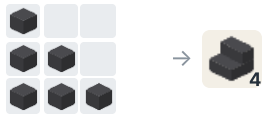
Polished deepslate
No table



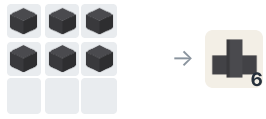
Polished deepslate
No table



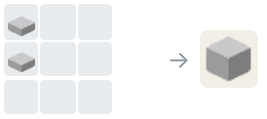
Polished deepslate slab
No table



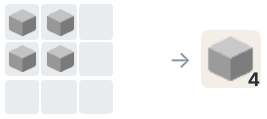
Polished deepslate stairs
No table



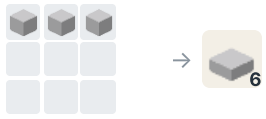
Polished deepslate wall
No table



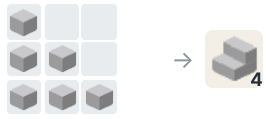
Polished diorite
No table



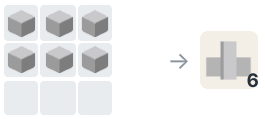
Polished diorite
No table



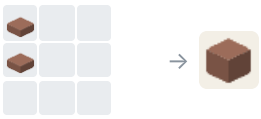
Polished diorite slab
No table



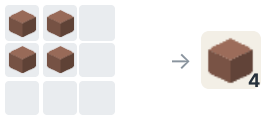
Polished diorite stairs
No table



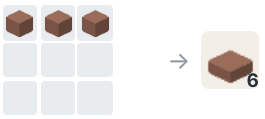
Polished diorite wall
No table



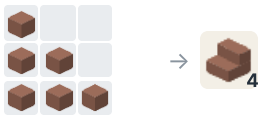
Polished granite
No table



Polished granite
No table



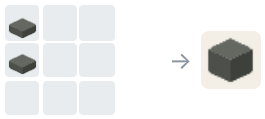
Polished granite slab
No table



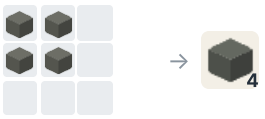
Polished granite stairs
No table



Polished granite wall
No table



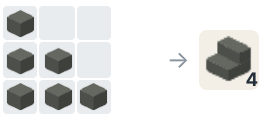
Polished tuff
No table



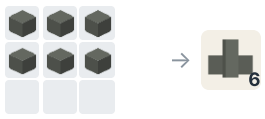
Polished tuff
No table



Polished tuff slab
No table



Polished tuff stairs
No table



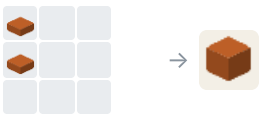
Polished tuff wall
No table



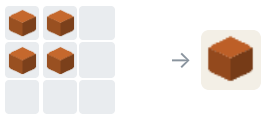
Purple fire glass
No table



Red fire glass
No table



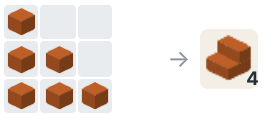
Red sandstone
No table



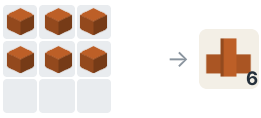
Red sandstone
No table



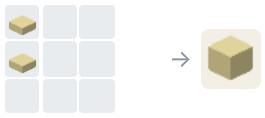
Red sandstone slab
No table



Red sandstone stairs
No table



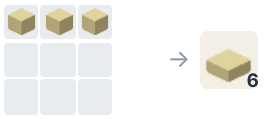
Red sandstone wall
No table



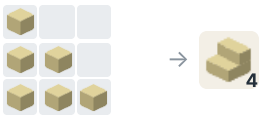
Sandstone
No table



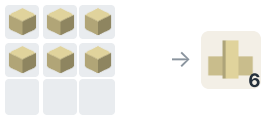
Sandstone
No table



Sandstone slab
No table



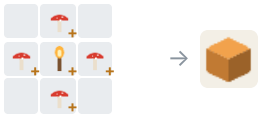
Sandstone stairs
No table



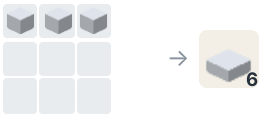
Sandstone wall
No table



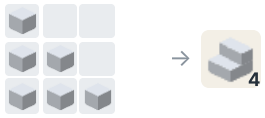
Sea lantern
No table



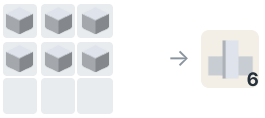
Shroomlight
No table



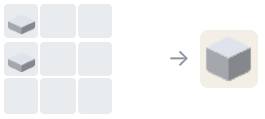
Silver brick slab
No table



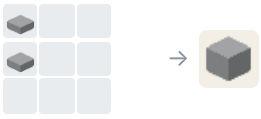
Silver brick stairs
No table



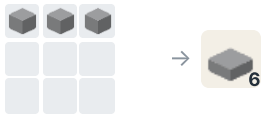
Silver brick wall
No table



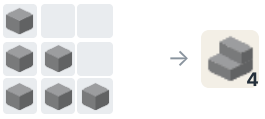
Silver bricks
No table



Smooth stone
No table



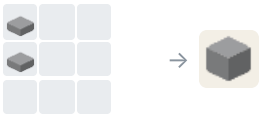
Smooth stone brick slab
No table



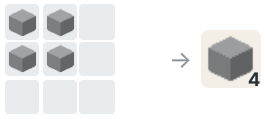
Smooth stone brick stairs
No table



Smooth stone brick wall
No table



Smooth stone bricks
No table



Smooth stone bricks
No table



Smooth stone slab
No table



Smooth stone stairs
No table



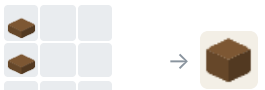
Smooth stone wall
No table



Spruce fence
No table



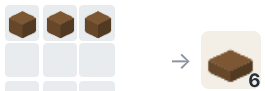
Spruce planks
No table



Spruce planks
No table



Spruce planks
No table



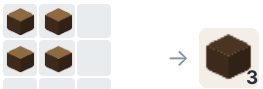
Spruce slab
No table



Spruce stairs
No table



Spruce wall
No table



Spruce wood
No table



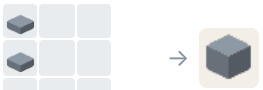
Steel brick slab
No table



Steel brick stairs
No table



Steel brick wall
No table



Steel bricks
No table



Stone
No table



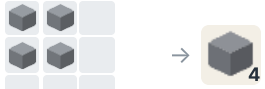
Stone brick slab
No table



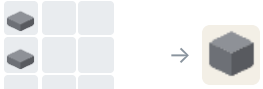
Stone brick stairs
No table



Stone brick wall
No table



Stone bricks
No table



Stone bricks
No table



Stone crafting table
No table



Stone slab
No table



Stone stairs
No table



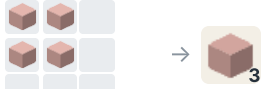
Stone wall
No table



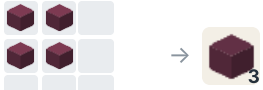
Stripped acacia wood
No table



Stripped birch wood
No table



Stripped cherry wood
No table



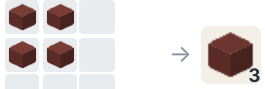
Stripped crimson hyphae
No table



Stripped dark oak wood
No table



Stripped jungle wood
No table



Stripped mangrove wood
No table



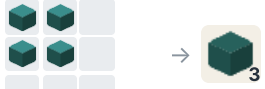
Stripped oak wood
No table



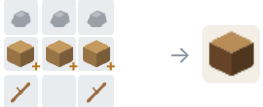
Stripped pale oak wood
No table



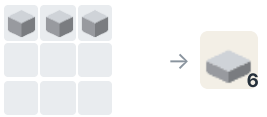
Stripped spruce wood
No table



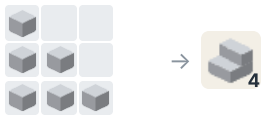
Stripped warped hyphae
No table



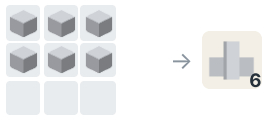
Tanning table
No table



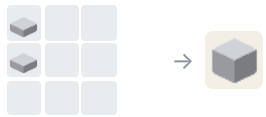
Tin brick slab
No table



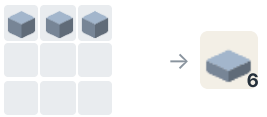
Tin brick stairs
No table



Tin brick wall
No table



Tin bricks
No table



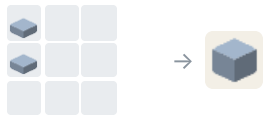
Titanium brick slab
No table



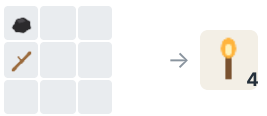
Titanium brick stairs
No table



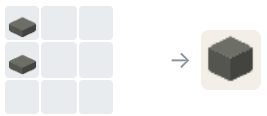
Titanium brick wall
No table



Titanium bricks
No table



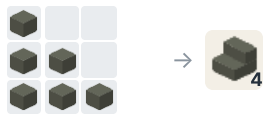
Torch
No table



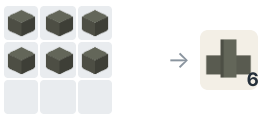
Tuff
No table



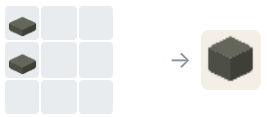
Tuff brick slab
No table



Tuff brick stairs
No table



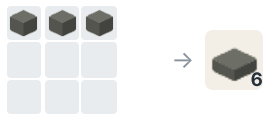
Tuff brick wall
No table



Tuff bricks
No table



Tuff bricks
No table



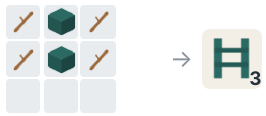
Tuff slab
No table



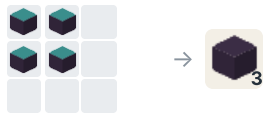
Tuff stairs
No table



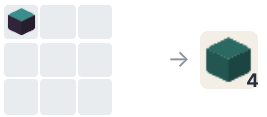
Tuff wall
No table



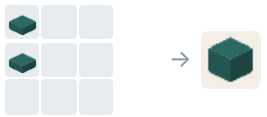
Warped fence
No table



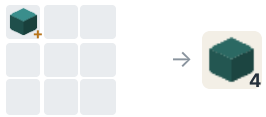
Warped hyphae
No table



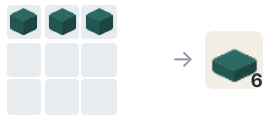
Warped planks
No table



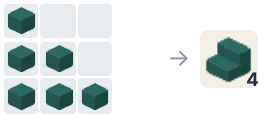
Warped planks
No table



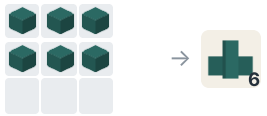
Warped planks
No table



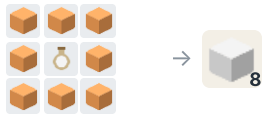
Warped slab
No table



Warped stairs
No table



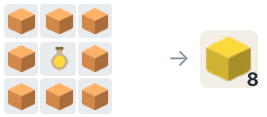
Warped wall
No table



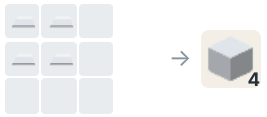
White fire glass
No table



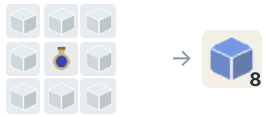
Witch's cauldron
No table



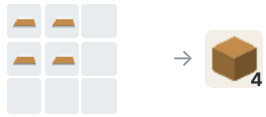
Yellow fire glass
No table



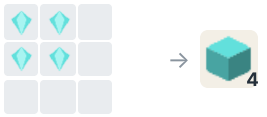
Aluminium bricks
Stone crafting table



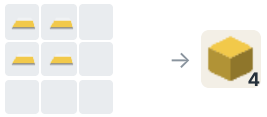
Blue glass
Stone crafting table



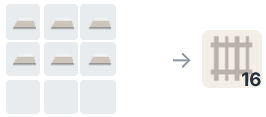
Bronze bricks
Stone crafting table



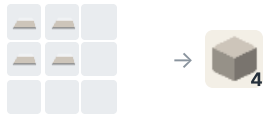
Diamond bricks
Stone crafting table



Gold bricks
Stone crafting table



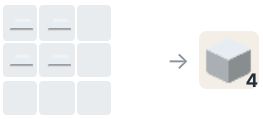
Iron bars
Stone crafting table



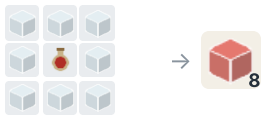
Iron bricks
Stone crafting table



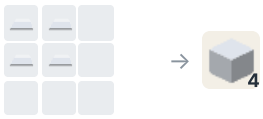
Lesser stone crucible
Stone crafting table



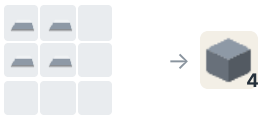
Platinum bricks
Stone crafting table



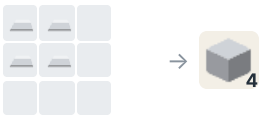
Red glass
Stone crafting table



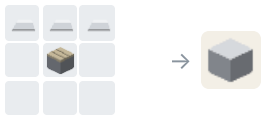
Silver bricks
Stone crafting table



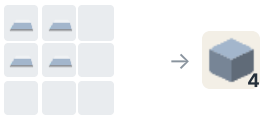
Steel bricks
Stone crafting table



Tin bricks
Stone crafting table



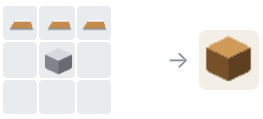
Tin crafting table
Stone crafting table



Titanium bricks
Stone crafting table



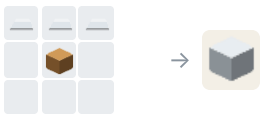
Yellow glass
Stone crafting table



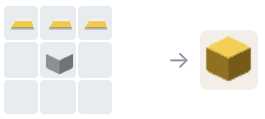
Bronze crafting table
Tin crafting table



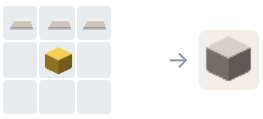
Bronze smelter
Tin crafting table



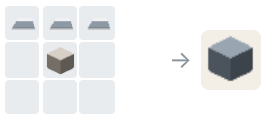
Aluminium crafting table
Bronze crafting table



Gold crafting table
Aluminium crafting table



Iron crafting table
Gold crafting table



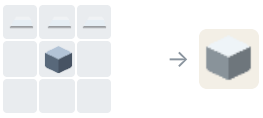
Steel crafting table
Iron crafting table



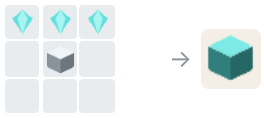
Steel forge
Iron crafting table



Titanium crafting table
Steel crafting table

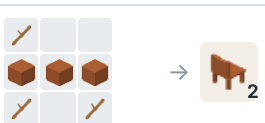


Platinum crafting table
Titanium crafting table

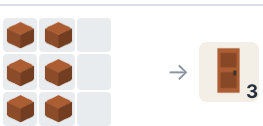


Diamond crafting table
Platinum crafting table

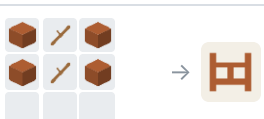
Decor (219 recipes)



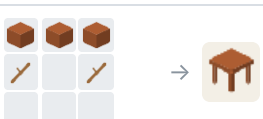
Acacia chair
No table



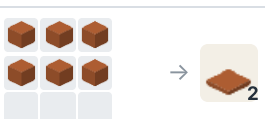
Acacia door
No table



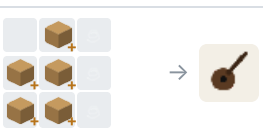
Acacia gate
No table



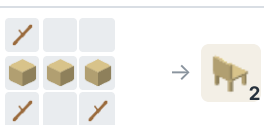
Acacia table
No table



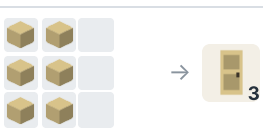
Acacia trapdoor
No table



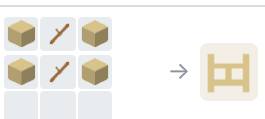
Bass
No table



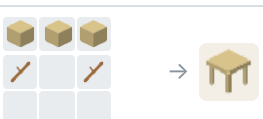
Birch chair
No table



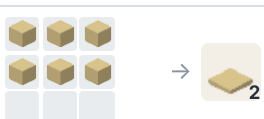
Birch door
No table



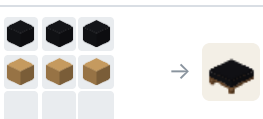
Birch gate
No table



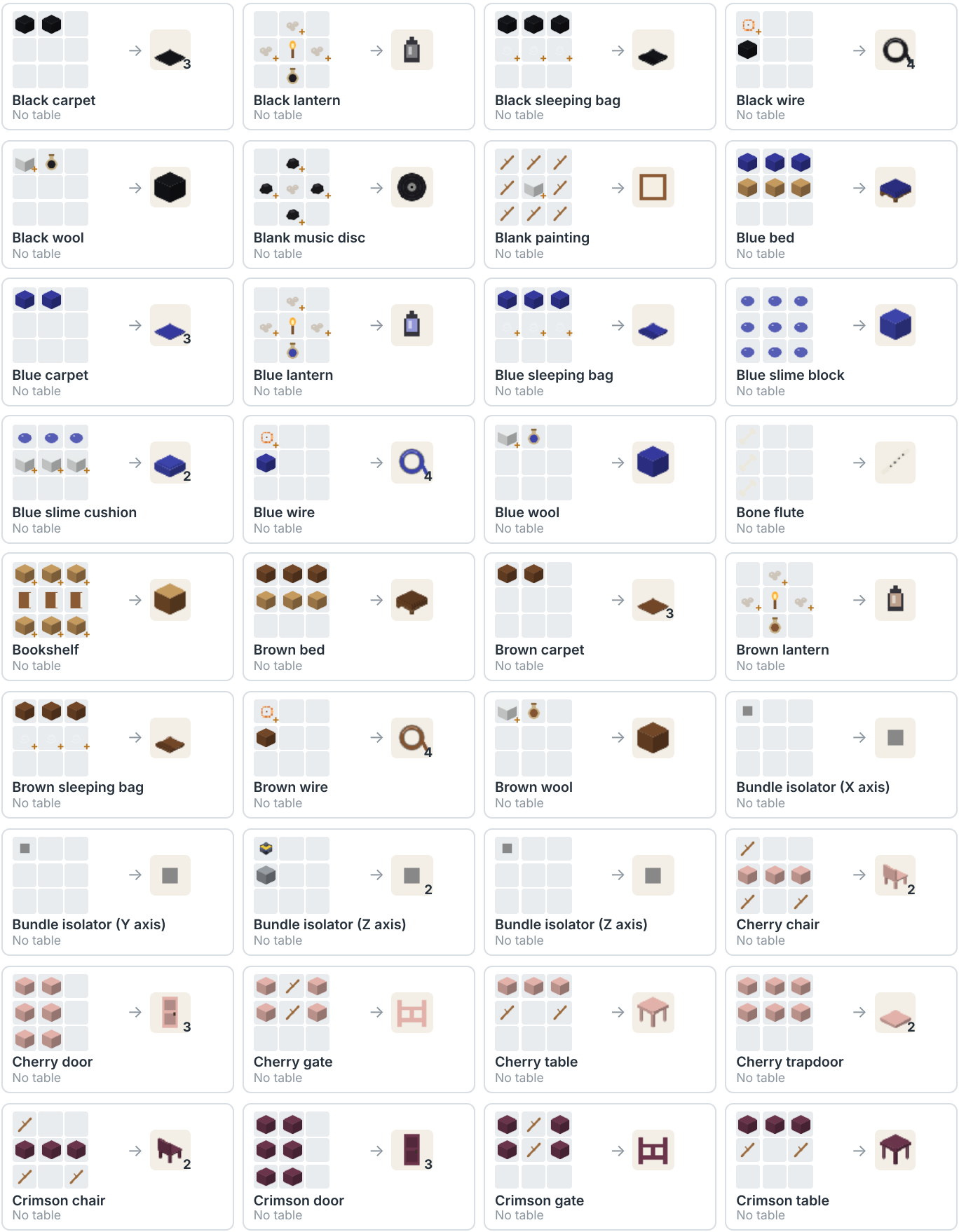
Birch table
No table

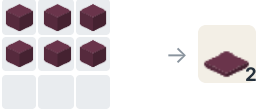


Birch trapdoor
No table



Black bed
No table


Black carpet
No table
Black lantern
No table
Black sleeping bag
No table
Black wire
No table
Black wool
No table
Blank music disc
No table
Blank painting
No table
Blue bed
No table
Blue carpet
No table
Blue lantern
No table
Blue sleeping bag
No table
Blue slime block
No table
Blue slime cushion
No table
Blue wire
No table
Blue wool
No table
Bone flute
No table
Bookshelf
No table
Brown bed
No table
Brown carpet
No table
Brown lantern
No table
Brown sleeping bag
No table
Brown wire
No table
Brown wool
No table
Bundle isolator (X axis)
No table
Bundle isolator (Y axis)
No table
Bundle isolator (Z axis)
No table
Bundle isolator (Z axis)
No table
Cherry chair
No table
Cherry door
No table
Cherry gate
No table
Cherry table
No table
Cherry trapdoor
No table
Crimson chair
No table
Crimson door
No table
Crimson gate
No table
Crimson table
No table



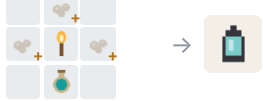
Crimson trapdoor
No table



Cyan bed
No table



Cyan carpet
No table



Cyan lantern
No table



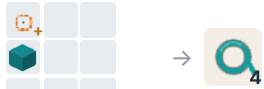
Cyan sleeping bag
No table



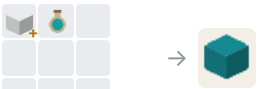
Cyan slime block
No table



Cyan slime cushion
No table



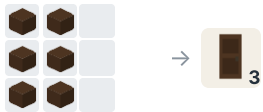
Cyan wire
No table



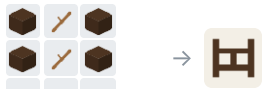
Cyan wool
No table



Dark oak chair
No table



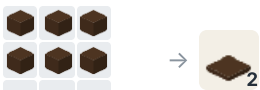
Dark oak door
No table



Dark oak gate
No table



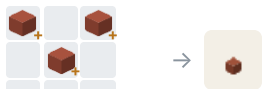
Dark oak table
No table



Dark oak trapdoor
No table



Drum kit
No table



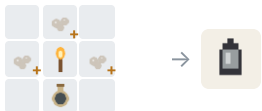
Flower pot
No table



Gray bed
No table



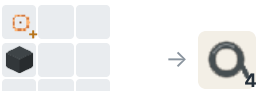
Gray carpet
No table



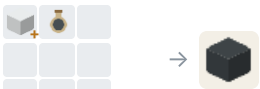
Gray lantern
No table



Gray sleeping bag
No table



Gray wire
No table



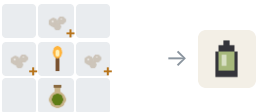
Gray wool
No table



Green bed
No table



Green carpet
No table



Green lantern
No table



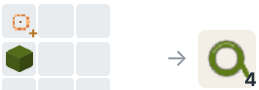
Green sleeping bag
No table



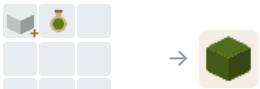
Green slime block
No table



Green slime cushion
No table



Green wire
No table



Green wool
No table



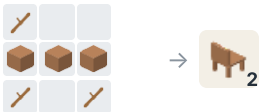
Hanging sign
No table



Item frame
No table



Jukebox
No table



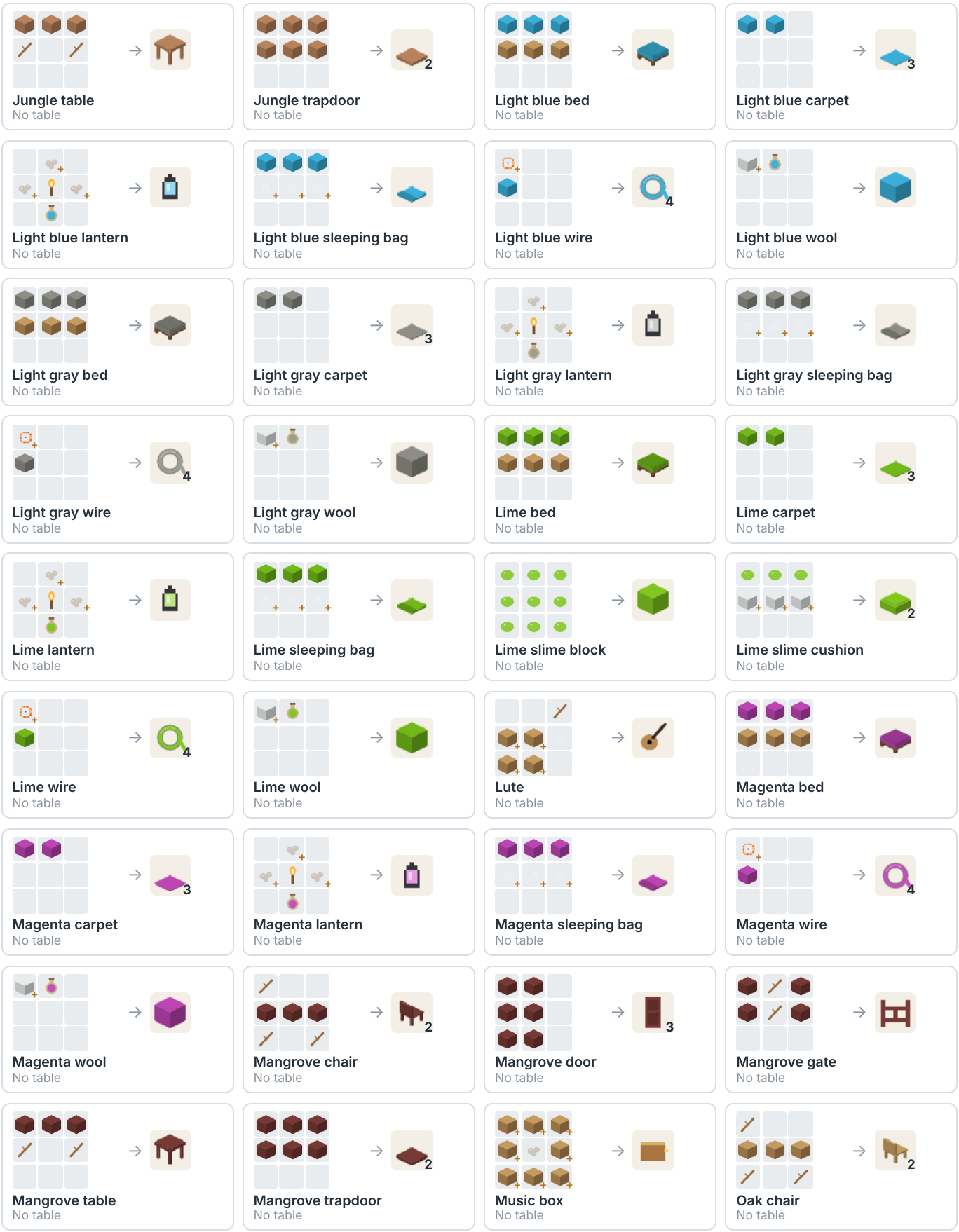
Jungle chair
No table



Jungle door
No table



Jungle gate
No table


Jungle table
No table
Jungle trapdoor
No table
2
Light blue bed
No table
Light blue carpet
No table
3
Light blue lantern
No table
Light blue sleeping bag
No table
Light blue wire
No table
4
Light blue wool
No table
Light gray bed
No table
Light gray carpet
No table
3
Light gray lantern
No table
Light gray sleeping bag
No table
Light gray wire
No table
4
Light gray wool
No table
Lime bed
No table
Lime carpet
No table
3
Lime lantern
No table
Lime sleeping bag
No table
Lime slime block
No table
Lime slime cushion
No table
2
Lime wire
No table
4
Lime wool
No table
Lute
No table
Magenta bed
No table
Magenta carpet
No table
3
Magenta lantern
No table
Magenta sleeping bag
No table
Magenta wire
No table
4
Magenta wool
No table
Mangrove chair
No table
2
Mangrove door
No table
3
Mangrove gate
No table
Mangrove table
No table
Mangrove trapdoor
No table
2
Music box
No table
Oak chair
No table
2



Oak door
No table

Oak gate
No table

Oak table
No table

Oak trapdoor
No table



Orange bed
No table

Orange carpet
No table

Orange lantern
No table

Orange sleeping bag
No table



Orange slime block
No table

Orange slime cushion
No table

Orange wire
No table

Orange wool
No table



Painting
No table

Pale oak chair
No table

Pale oak door
No table

Pale oak gate
No table

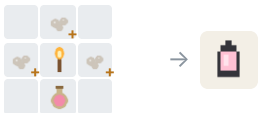


Pale oak table
No table

Pale oak trapdoor
No table

Pink bed
No table

Pink carpet
No table



Pink lantern
No table

Pink sleeping bag
No table

Pink slime block
No table

Pink slime cushion
No table



Pink wire
No table

Pink wool
No table

Poster
No table

Purple bed
No table

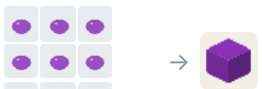


Purple carpet
No table

Purple lantern
No table

Purple lantern
No table

Purple sleeping bag
No table

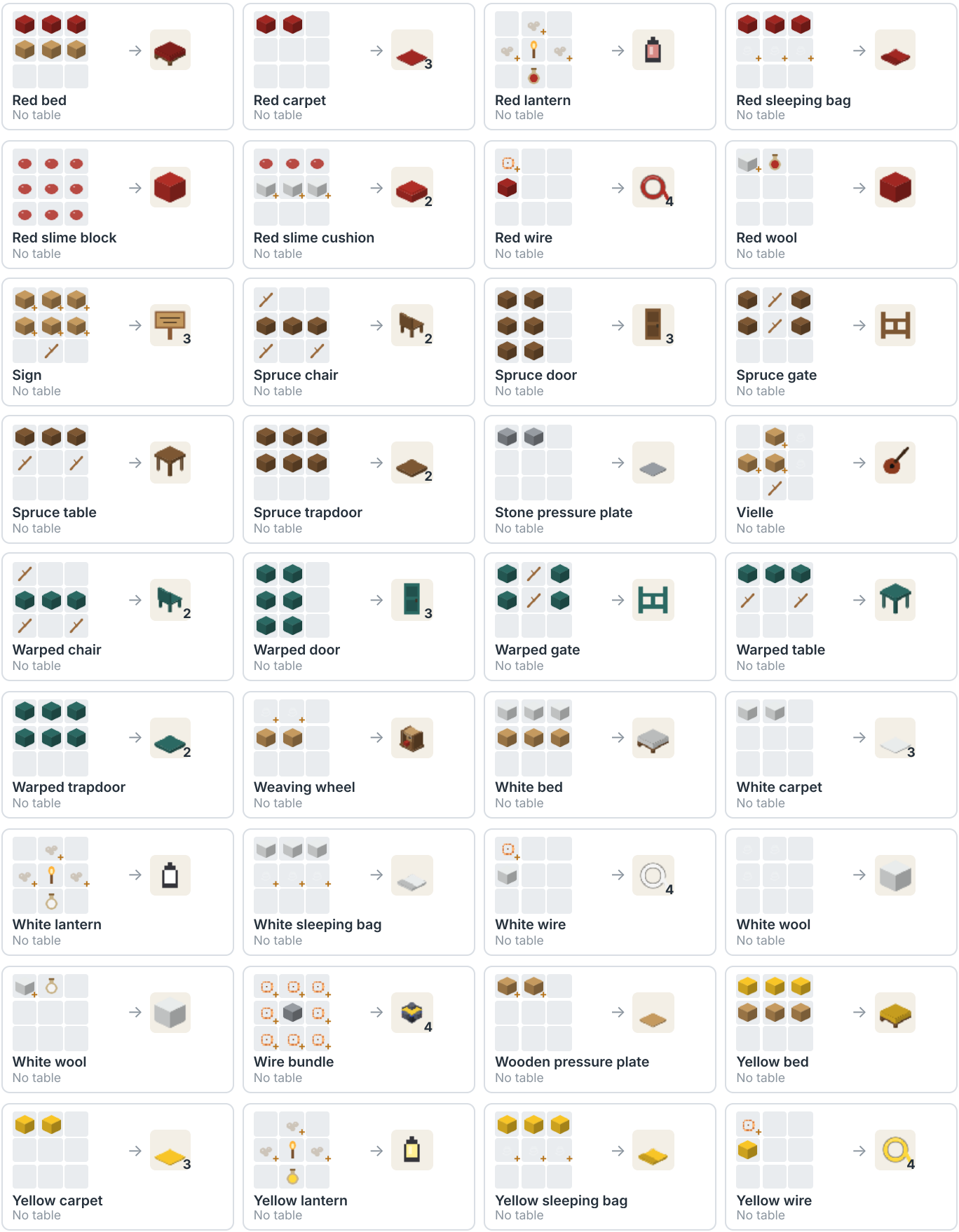


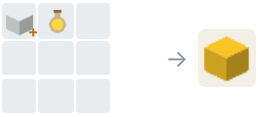
Purple slime block
No table

Purple slime cushion
No table

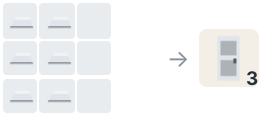
Purple wire
No table

Purple wool
No table

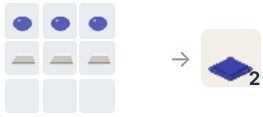

Red bed
No table
Red carpet
No table
3
Red lantern
No table
Red sleeping bag
No table
Red slime block
No table
Red slime cushion
No table
2
Red wire
No table
4
Red wool
No table
Sign
No table
3
Spruce chair
No table
2
Spruce door
No table
3
Spruce gate
No table
Spruce table
No table
Spruce trapdoor
No table
2
Stone pressure plate
No table
Vielle
No table
Warped chair
No table
2
Warped door
No table
3
Warped gate
No table
Warped table
No table
Warped trapdoor
No table
2
Weaving wheel
No table
White bed
No table
White carpet
No table
3
White lantern
No table
White sleeping bag
No table
White wire
No table
4
White wool
No table
White wool
No table
Wire bundle
No table
4
Wooden pressure plate
No table
Yellow bed
No table
Yellow carpet
No table
3
Yellow lantern
No table
Yellow sleeping bag
No table
Yellow wire
No table
4



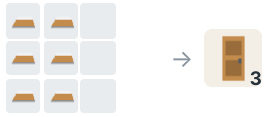
Yellow wool
No table



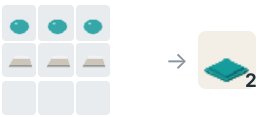
Aluminium door
Stone crafting table



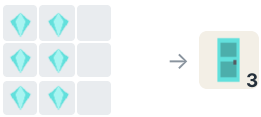
Blue bounce pad
Stone crafting table



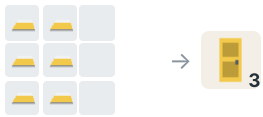
Bronze door
Stone crafting table



Cyan bounce pad
Stone crafting table



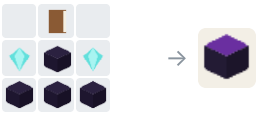
Diamond door
Stone crafting table



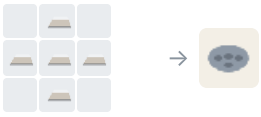
Gold door
Stone crafting table



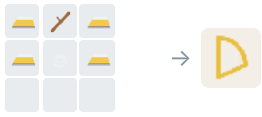
Green bounce pad
Stone crafting table



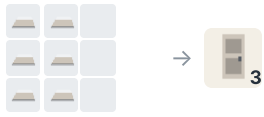
Grimoire desk
Stone crafting table



Handpan
Stone crafting table



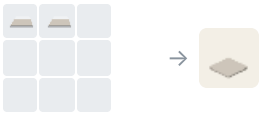
Harp
Stone crafting table



Iron door
Stone crafting table



Iron gate
Stone crafting table



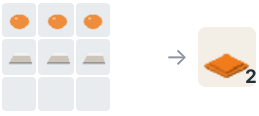
Iron pressure plate
Stone crafting table



Iron trapdoor
Stone crafting table



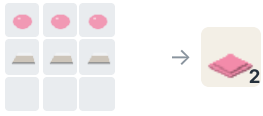
Lime bounce pad
Stone crafting table



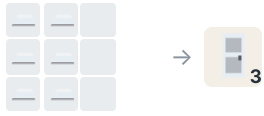
Orange bounce pad
Stone crafting table



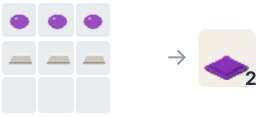
Piano
Stone crafting table



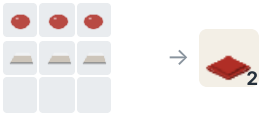
Pink bounce pad
Stone crafting table



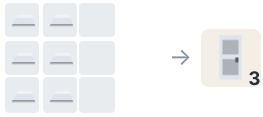
Platinum door
Stone crafting table



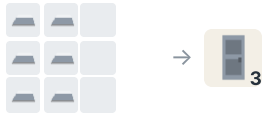
Purple bounce pad
Stone crafting table



Red bounce pad
Stone crafting table



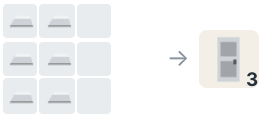
Silver door
Stone crafting table



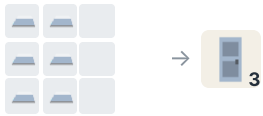
Steel door
Stone crafting table



Synth
Stone crafting table



Tin door
Stone crafting table

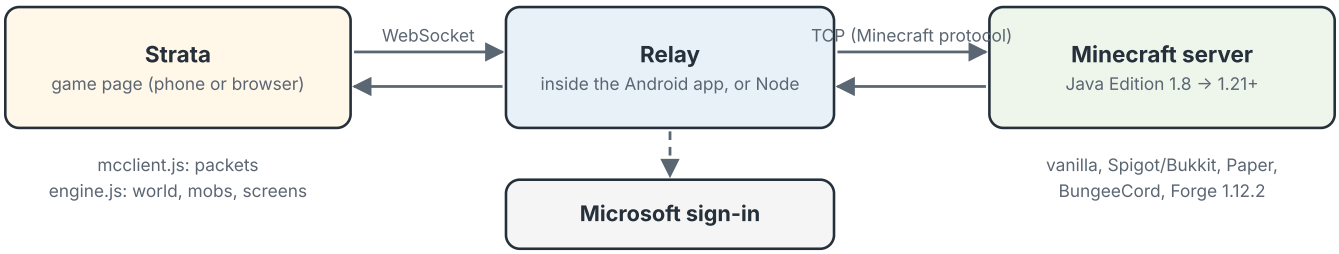


Titanium door
Stone crafting table

Minecraft server compatibility

Multiplayer → Minecraft Compat lets Strata join Minecraft: Java Edition servers. Strata speaks the network protocol itself and draws the server's world with its own engine. No Minecraft code is used.

How a connection works



A web page can't open the raw TCP connection Minecraft uses, so Strata talks to a small **relay** over a WebSocket, and the relay forwards bytes to the server. The Android app runs the relay inside itself, so nothing extra is needed on a phone. In a desktop browser, run `tools/relay/relay.mjs` with Node.js and enter its address (by default `ws://localhost:25580`).

> **Accounts.** Online-mode servers need a Microsoft sign-in. Strata's relay has a "Sign in with Microsoft" device-code flow (your own Azure app id is needed for the desktop relay); only your name and UUID reach the page, and tokens stay in the relay. This sign-in has been tested against stand-in endpoints only so far. Offline-mode servers (where the owner has turned off account checks) have been tested in practice. Please only play on servers you're allowed to play on.

Version table



Minecraft versions	Protocol	Login and chat	Full world and gameplay	Forge mods
1.8 – 1.8.9	47	Yes	Yes the first version built, the most polished	No
1.9 – 1.9.4	107 – 110	Yes	Yes off hand, elytra, shields	No

Minecraft versions	Protocol	Login and chat	Full world and gameplay	Forge mods
1.10 – 1.10.2	210	Yes	Yes	No
1.11 – 1.11.2	315 – 316	Yes	Yes shulkers, llamas, illagers	No
1.12 – 1.12.2	335 – 340	Yes	Yes recipe book, advancements, bed colours	Yes 1.12.2 (Forge 14.23.x)
1.13 – 1.21.x and later	393 – 775	Yes sign-in, chat, commands	Not yet the 1.13 block-ID change needs its own world code	No

Every minor release is listed by name in the version menu; releases sharing a protocol number behave the same on the wire. Packet tables cover 48 protocol numbers.

What works in-game (1.8 – 1.12.2)

- Chunks, block changes, lighting, signs and heads; other players with skins, armour, held items and poses.
- Mobs with Minecraft-shaped models (resource-pack textures when you have a pack), drops, projectiles, riding boats and animals.
- Mining with real timing, placing, inventories, chests, furnaces, crafting tables, enchanting, anvils, brewing, villager trades, the 1.12 recipe book.
- Health, hunger, eating and drinking with animations, XP, potion effects, armour, elytra gliding and firework boosts, shields.
- Plugin features: holograms, NPC name tags, team colours, clickable chat commands and hover text, tab completion, scoreboards, boss bars, titles, action bars.
- Server resource packs (downloaded and layered over yours for the session), maps in hand and in item frames, world borders, cooldowns, advancements toasts, BungeeCord/Velocity server switches.
- Sounds by id: from your pack when it has them, otherwise Strata's own synthesized sounds.
- **Skipped:** anything Strata has no model or system for is ignored rather than shown wrong.



A 1.12.2 test server: other players, mobs, signs and a boss bar.



Minecraft mobs drawn on Strata's models with resource-pack textures.

Forge 1.12.2 mode and mod compatibility

Turn on **Forge server (1.12.2 mods)** in the connect screen to join modded servers. Strata answers Forge's handshake, learns the server's block and item registries, and runs small Strata-native "ports" of supported mods' client sides.

Joining a Forge server



1. On the connect screen, choose 1.12.2, tick **Forge server**, and import the server's mod jars (Chapter 9). Jars only give Strata the mods' textures, names and models; they never run.
2. Strata reads the server's mod list from its status ping and joins with Forge's marker. The FML handshake exchanges mod lists and the registries that say which number means which block or item.
3. In the world, modded blocks and items use the textures from your jars (or plain colours with readable names without them). Mod screens with item slots open as normal windows; ported mods add their special parts.

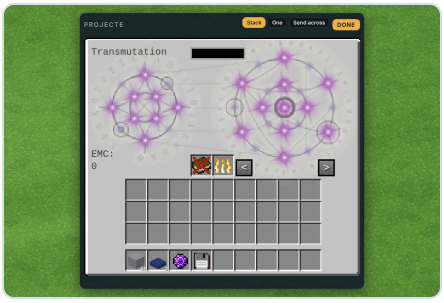
Compatibility table

Ported tested on a real Forge 1.12.2 server · **Generic** blocks, items and slot screens work without a port · **Partial** works, with gaps noted · **Not yet** not supported

Mod	Version tested	Status	What works	Gaps and notes
ProjectE Equivalent Exchange	1.12.2-PE1.4.1	Ported	Transmutation Table and Tablet laid out like the mod's screen (learning, EMC, search, pages, take one or a stack), EMC values on items, condenser, collector and relay readouts, the mod's key buttons (Armor, Charge, Extra, Fire, Mode), condensers and alchemical chests drawn from their models.	Client-only visual effects of some tools and rings aren't drawn.
CC: Tweaked ComputerCraft	1.89.2	Ported	Computer, turtle and pocket computer terminals with typing, keys, mouse, paste, terminate/reboot/shutdown; monitors drawn in the world (any facing); turtles drawn from their model with upgrades and dye, moving smoothly.	Printed pages aren't viewable yet; speakers play through server sounds.
Solar Flux Reborn	4.35r	Ported	Panels drawn at their real height and textures; the panel screen shows energy, generation, maximum and sunlight.	—
Waystones	4.1.0	Ported	Waystones drawn from their model with the glowing rune once activated; naming, the destination list from stones, warp stones and warp scrolls; teleport effect and sound.	—
Warp Book	1.12.2-3.3.4	Ported	The book's page list and warping, binding pages by name, bound pages warp on use, page names and tinted icons, arrive and depart sounds.	Page inventory, drinking warp potions, Teleporter and Book Cloner blocks are wired but untested.
Ender Mail	1.12.2-1.1.3	Ported	Stamping packages (coordinates or locker ID), calling the Ender Mailman with a Package Controller, delivery status on the controller, naming lockers, the Mailman drawn from the mod's model.	The package the Mailman carries isn't drawn in his hands.
Packing Tape	0.7.6	Ported	Crouch and tap a chest with tape to pack it; place and unpack the package.	Needs no port beyond crouch reaching the server.
Elevator Mod	1.12.2-1.3.13	Ported	Jump to go up, crouch to go down, following the server's range and same-colour settings; the mod's teleport sound.	Uses Strata's own elevator controls.
Any other 1.12.2 mod	—	Generic	Blocks and items with the jar's textures, names and box shapes; container screens with slots; modded creatures with humanoid textures on a humanoid body; modded sounds the server sends.	Custom screens, buttons and rendering need a port. Other creatures use a stand-in body.

Mod	Version tested	Status	What works	Gaps and notes
Botania, EnderIO, RFTools, OpenBlocks	—	Wanted	Generic support only for now.	Planned ports.
RLCraft (modpack)	—	Not yet	—	Too many client-side mods for now.

Mod by mod



ProjectE's Transmutation Table, laid out like the mod's own screen.



A CC: Tweaked computer running CraftOS, with the on-screen keys.



ProjectE condensers and alchemical chests drawn from their models.



Waystones drawn from the mod's model, rune glowing once activated.



Solar Flux panels and modded blocks with textures from the imported jars.



The Ender Mailman (bottom left) in his postman's cap.

> **How a port works.** A port is a small piece of Strata code keyed by the mod's id. It can draw blocks the mod draws in code, read the mod's network messages, add to its screens, and open screens the mod opens on the client. It never runs the mod's own Java code. Chapter 11 goes deeper.

Importing mods, resource packs and skins

Everything you bring in stays on your device. Strata never uploads your files.



Forge mod jars

1. Get the same mod jars the server uses (the exact versions matter). Server owners usually share them as a modpack or a folder.
2. Multiplayer → Minecraft Compat → choose **1.12.2** → tick **Forge server (1.12.2 mods)**.
3. Tap **Import mod jars...** and pick the `.jar` files, or one `.zip` of the whole mods folder. You can import more later; **Clear** removes them all.
4. Wait for "Imported ...". The first import of many jars can take a minute on a phone. Strata keeps each mod's block and item textures, models and shapes, creature textures, screen backgrounds, names and sounds.
5. Join the server. The mod textures are layered over your resource pack for that session.

> If you update a mod or Strata adds support for more of a mod's files (sounds, screens), import its jar again.

Minecraft resource packs

1. Open a world (single player or a server), pause, and find **Resource pack**.
2. Tap **Import .zip** and pick a Java Edition resource pack. Old (1.8–1.12, `textures/blocks`) and new (1.13+, `textures/block`) layouts both work.
3. Strata uses the textures for blocks and items it has (animated water and lava too), creature skins, armour, paintings, the destroy stages and up to about 40 MB of sounds. **Remove** goes back to the default look.

In single player, a pack belongs to that world. On a server, it lasts for the session; a server's own pack is layered on top, and mod textures sit on top of both.

Player skins

Pause → **Import skin .png**. Any Java skin (64×64 or the old 64×32) works; both layers are drawn and slim arms are detected, or choose Classic or Slim yourself. On servers, other players' skins come from their profiles.

Building a Minecraft resource pack yourself

Strata can't include Minecraft's textures and sounds. Instead it comes with a small script that fetches them from Mojang's own download servers, the same ones the official launcher uses, and builds a resource pack on your computer.



every download is checked against Mojang's published SHA-1 · cached for re-runs · no game code kept

Running it

1. Install Python 3.8 or newer (python.org). Nothing else is needed.
2. Save `tools/make_resource_pack.py` from the Strata repository (the full script is in Appendix A).
3. In a terminal, in the folder you saved it to, run one of:

```
python3 make_resource_pack.py                  # the latest release
python3 make_resource_pack.py --version 1.12.2  # match a 1.12.2 server
python3 make_resource_pack.py --list            # every release you can pick
python3 make_resource_pack.py --music           # also music and discs (large)
python3 make_resource_pack.py --no-sounds       # textures only
```

4. It prints progress and finishes with `Minecraft-<version>-pack.zip`. Downloads are cached in `.mcpack-cache`, so running it again is quick.
5. Copy the zip to your phone if needed, then import it in Strata (Pause → Resource pack → Import .zip).

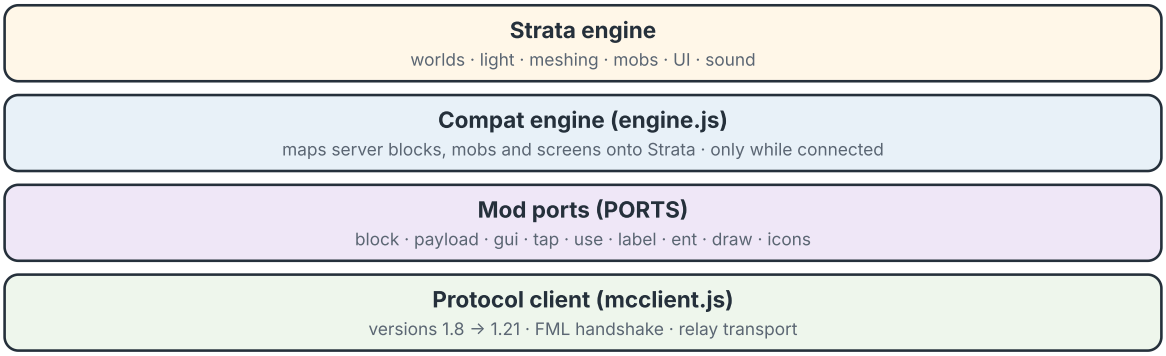
What it fetches

- The version list and the chosen version's file list from `piston-meta.mojang.com`.
- The game's client file, from which it copies only `assets/minecraft/textures`, `models`, `blockstates` and `pack.png`. No game code is kept.
- Sounds from `resources.download.minecraft.net`, listed by the version's asset index, plus `sounds.json`. Music and discs are left out unless you ask. The sounds Strata uses most (digging, footsteps, blocks, creatures) are written first, because Strata reads about the first 40 MB of a pack's sounds.
- Every file is checked against the checksum Mojang publishes for it.

Keep the pack to yourself. The files belong to Mojang. You should own Minecraft: Java Edition, and the pack is for your own use. Don't share, upload or sell it. That's why Strata ships the script and not the files.

Under the hood: the technical complexities

Some of the harder problems behind Strata and its compatibility layer, for the curious and for anyone who wants to experiment with the code.



One file, one engine

The whole game is one HTML file (`src/strata.html`) using WebGL2, with the Minecraft client (`src/mc/mcclient.js`) and compatibility engine (`src/mc/engine.js`) embedded by a build script. A tiny release workflow turns every push into a standalone file and an Android app.

Light that only updates where it changes

Coloured block light and sky light are stored per chunk. A new chunk is lit once; after that, only edits to light-relevant blocks re-solve light in a small queue around them, which keeps big worlds smooth on phones.

The isolation rule

Anything added for Minecraft exists only while connected. Compat blocks use ids 8192–65535 (Strata uses up to 6240), are registered when a session starts and removed when it ends, and never touch saves, recipes, world generation or the creative list.

Climbing the protocol ladder

Minecraft's network format changed in every version. Strata built 1.8 first, then added 1.9's paletted chunk sections, UUIDs and fixed-point movement, 1.9's entity metadata (mapped back to the meanings the 1.8 code uses), 1.10's metadata shift and 1.11's varint mob types, up to 1.12.2. Minecraft block states far outnumber Strata's block ids, so states are mapped to compat ids on demand.

Forge: registries instead of fixed numbers

On a Forge server, block and item numbers are handed out per server. Strata reads the registries from the FML handshake and identifies everything by its registry name (`projecte:condenser_mk1`), so a mod's block is recognised however it was numbered.

Ports do what the mod's client half would

Many mods decide things on the client: ProjectE works out the Transmutation Table's outputs itself, Waystones and Warp Book open their screens without the server, the Elevator mod searches for the next

elevator, and Ender Mail stamps from a client screen. Each port reproduces exactly that half, reading the mod's own network messages byte for byte. Some server behaviour also needs care: Ender Mail stamps from its network thread without sending the block change, so Strata shows the package stamped itself.

Drawing mod models

Mod blocks come from their JSON models (boxes with per-face textures). Things mods draw in code (turtles, waystones, condensers, the Ender Mailman) are rebuilt from the mod's model boxes, read out of its compiled classes, with the textures from your jar.

Sounds without shipping sounds

Server sounds arrive as numbers. Vanilla numbers map through per-version tables; modded ones through Forge's sound registry to the mod's `sounds.json` and its own `.ogg` files from your jar. Sounds a mod plays only on the client (Warp Book's warp, ProjectE's charge) are played by the port.

Telling the server you're sneaking

Many mods use "sneak + right click" to mean "use the item instead of opening the block". Strata sends the sneak state whenever you crouch, so packing tape, page inventories and renaming work.

Credits

Strata

ToshieCosed: creator, designer and author of Strata, its engine, its tech and magic trees and its Minecraft and Forge compatibility layer. Source and releases: github.com/ToshieCosed/MeadowMagic.

Mod authors

Strata's ports are written from scratch to talk to these mods. The mods, their designs, names, textures, models and sounds belong to their authors. Please support them on their own pages.

Mod	Authors	Notes and links
ProjectE	sinkillerj, Moze_Intel and contributors	Based on Equivalent Exchange 2 by x3n0ph0b3, whose assets it uses with permission. curseforge.com/minecraft/mc-mods/projecte
CC: Tweaked	SquidDev; ComputerCraft by Daniel Ratcliffe (dan200) and Aaron Mills	github.com/SquidDev-CC/CC-Tweaked
Solar Flux Reborn	Zeitheron	curseforge.com/minecraft/mc-mods/solar-flux-reborn
Waystones	BlayTheNinth	Includes item icons by JoeCreates (CC BY-SA 3.0). curseforge.com/minecraft/mc-mods/waystones
Warp Book	ferreusveritas; original by panicnot42	1.8 port by NightKosh, plus the translators credited in the mod. curseforge.com/minecraft/mc-mods/warp-book
Ender Mail	ChaosTheDude (Matt Czyr)	github.com/MattCzyr/EnderMail
Packing Tape	gigaherz	github.com/gigaherz/PackingTape
Elevator Mod	VsnGamer	Based on the OpenBlocks elevator by the OpenBlocks team. curseforge.com/minecraft/mc-mods/openblocks-elevator

Data, tools and fonts

- **PrismarineJS minecraft-data** (MIT licence): packet ids, item and block tables, recipes and sound-event lists for each version, generated into Strata at build time. **node-minecraft-protocol** and **prismarine-chunk** are used only in Strata's development tests.
- **Minecraft Forge**: the server Strata's Forge mode is tested against.
- Player skins on servers are loaded from Mojang's skin service and Crafatar.
- Interface fonts from Google Fonts (SIL Open Font Licence).

Licence and legal notes

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```
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```

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- **No Minecraft or mod assets are distributed.** Textures, sounds and models come only from files you provide (your resource packs and mod jars) or make yourself with the script in Chapter 10, from your own copy of the game.
- **Mods belong to their authors** (see Credits). Mod names are used only to say what Strata works with.
- If you believe something in Strata infringes your rights, open an issue on the GitHub repository and it will be looked at promptly.

This guide explains the project's intent and licence in plain language; it isn't legal advice.

The resource pack script

Saved in the repository as `tools/make_resource_pack.py`. Standard-library Python only; it downloads only from Mojang's servers.

```
#!/usr/bin/env python3
"""Build a Minecraft Java resource pack from your own copy of the game's files.

Strata cannot ship Minecraft's textures or sounds. This script fetches them
from Mojang's official download servers, the same ones the Minecraft
launcher uses, and packs them into a resource pack .zip on your computer.
Then import that pack in Strata from the pause menu.

  python3 make_resource_pack.py                 # latest release
  python3 make_resource_pack.py --version 1.12.2 # a specific version
  python3 make_resource_pack.py --list          # list available versions

Options:
  --version V     Minecraft version to fetch (default: latest release)
  --out FILE      output zip (default: Minecraft-<version>-pack.zip)
  --music         also include music and music discs (very large)
  --no-sounds     textures only
  --cache DIR     where downloads are kept between runs (default: .mcpack-cache)

Needs Python 3.8 or newer and nothing else. Run it yourself, for your own
use: you should own Minecraft: Java Edition, and the pack it makes is for you
alone. Don't share or upload the zip, because its contents belong to Mojang.

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Strata (PolyForm Noncommercial 1.0.0).
"""
import argparse, concurrent.futures, hashlib, io, json, os, sys, time, urllib.request, zipfile

META = os.environ.get('MCPACK_META', 'https://piston-
meta.mojang.com/mc/game/version_manifest_v2.json')
RES = os.environ.get('MCPACK_RES', 'https://resources.download.minecraft.net')
UA = 'StrataPackBuilder/1.0'

# Strata reads about 40 MB of a pack's sounds, in the order they sit in the zip,
# so the sounds it uses most go in first.
SOUND_ORDER = ['dig/', 'step/', 'block/', 'random/', 'damage/', 'mob/', 'entity/', 'liquid/',
'fire/',
               'item/', 'note/', 'ambient/', 'portal/', 'minecart/', 'tile/', 'enchant/',
'fireworks/', 'game/']


def fetch(url, sha1=None, cache=None, tries=4):
    """Download url, checked against sha1 when given, using the cache folder when possible."""
    path = None
    if cache and sha1:
        path = os.path.join(cache, sha1[:2], sha1)
        if os.path.exists(path):
            with open(path, 'rb') as f:
                data = f.read()
            if hashlib.sha1(data).hexdigest() == sha1:
                return data
```

```
    for n in range(tries):
        try:
            req = urllib.request.Request(url, headers={'User-Agent': UA})
            with urllib.request.urlopen(req, timeout=60) as r:
                data = r.read()
            if sha1 and hashlib.sha1(data).hexdigest() != sha1:
                raise IOError('checksum mismatch for ' + url)
            if path:
                os.makedirs(os.path.dirname(path), exist_ok=True)
                with open(path, 'wb') as f:
                    f.write(data)
            return data
        except Exception as e:  # network hiccups: wait and retry
            if n == tries - 1:
                raise
            time.sleep(1.5 * (n + 1))


def sound_rank(name):
    for i, p in enumerate(SOUND_ORDER):
        if name.startswith(p):
            return i
    return len(SOUND_ORDER)


def main():
    ap = argparse.ArgumentParser(description='Build a Minecraft resource pack from Mojang\'s
download servers.')
    ap.add_argument('--version')
    ap.add_argument('--out')
    ap.add_argument('--music', action='store_true')
    ap.add_argument('--no-sounds', action='store_true')
    ap.add_argument('--list', action='store_true')
    ap.add_argument('--cache', default='.mcpack-cache')
    a = ap.parse_args()

    print('Reading the version list...')
    manifest = json.loads(fetch(META))
    if a.list:
        for v in manifest['versions']:
            if v['type'] == 'release':
                print(v['id'])
        return
    vid = a.version or manifest['latest']['release']
    entry = next((v for v in manifest['versions'] if v['id'] == vid), None)
    if not entry:
        sys.exit('No version called %s. Use --list to see them.' % vid)
    ver = json.loads(fetch(entry['url'], entry.get('sha1'), a.cache))
    out = a.out or 'Minecraft-%s-pack.zip' % vid

    cj = ver['downloads']['client']
    print('Downloading the %s game files (%.1f MB)...' % (vid, cj['size'] / 1e6))
    client = zipfile.ZipFile(io.BytesIO(fetch(cj['url'], cj['sha1'], a.cache)))

    # pack.mcmeta: the version's own pack format when it says one, else a sensible guess
    fmt = 3 if vid.startswith('1.12') or vid.startswith('1.11') else 34
    try:
        vj = json.loads(client.read('version.json'))
        pv = vj.get('pack_version')
```

```
        fmt = pv.get('resource', fmt) if isinstance(pv, dict) else (pv or fmt)
    except KeyError:
        pass
    if vid.startswith(('1.8', '1.9', '1.10')):
        fmt = 1 if vid.startswith('1.8') else 2

    with zipfile.ZipFile(out, 'w', zipfile.ZIP_DEFLATED) as z:
        z.writestr('pack.mcmeta', json.dumps({'pack': {'pack_format': fmt, 'description':
'Minecraft %s (made on this computer for personal use)' % vid}}, indent=2))
        n = 0
        for info in client.infolist():
            name = info.filename
            if name.startswith(('assets/minecraft/textures/', 'assets/minecraft/blockstates/',
'assets/minecraft/models/')) or name == 'pack.png':
                z.writestr(name, client.read(name))
                n += 1
        print('  %d textures and models' % n)

        if not a.no_sounds:
            ai = ver['assetIndex']
            index = json.loads(fetch(ai['url'], ai['sha1'], a.cache))['objects']
            if 'minecraft/sounds.json' in index:
                h = index['minecraft/sounds.json']['hash']
                z.writestr('assets/minecraft/sounds.json', fetch('%s/%s/%s' % (RES, h[:2], h),
h, a.cache))
            want = []
            for key, o in index.items():
                if not (key.startswith('minecraft/sounds/') and key.endswith('.ogg')):
                    continue
                rel = key[len('minecraft/sounds/'):]
                if not a.music and rel.startswith(('music/', 'records/')):
                    continue
                want.append((sound_rank(rel), rel, o['hash'], o['size']))
            want.sort()
            total = sum(w[3] for w in want)
            print('Downloading %d sounds (%.1f MB)...' % (len(want), total / 1e6))
            got, done = {}, 0
            with concurrent.futures.ThreadPoolExecutor(8) as ex:
                futs = {ex.submit(fetch, '%s/%s/%s' % (RES, h[:2], h), h, a.cache): rel for _,
rel, h, _ in want}
                for f in concurrent.futures.as_completed(futs):
                    got[futs[f]] = f.result()
                    done += 1
                    if done % 100 == 0 or done == len(want):
                        print('  %d / %d' % (done, len(want)), end='\r', flush=True)
            print()
            for _, rel, _, _ in want:  # written in priority order
                z.writestr('assets/minecraft/sounds/' + rel, got[rel])

    print('Done: %s (%.1f MB)' % (out, os.path.getsize(out) / 1e6))
    print('Import it in Strata from the pause menu (Resource pack, Import .zip). Keep it to
yourself: the files belong to Mojang.')


if __name__ == '__main__':
    main()
```

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